

**Andrew Kieckhefer**

*Curriculum Vitae*

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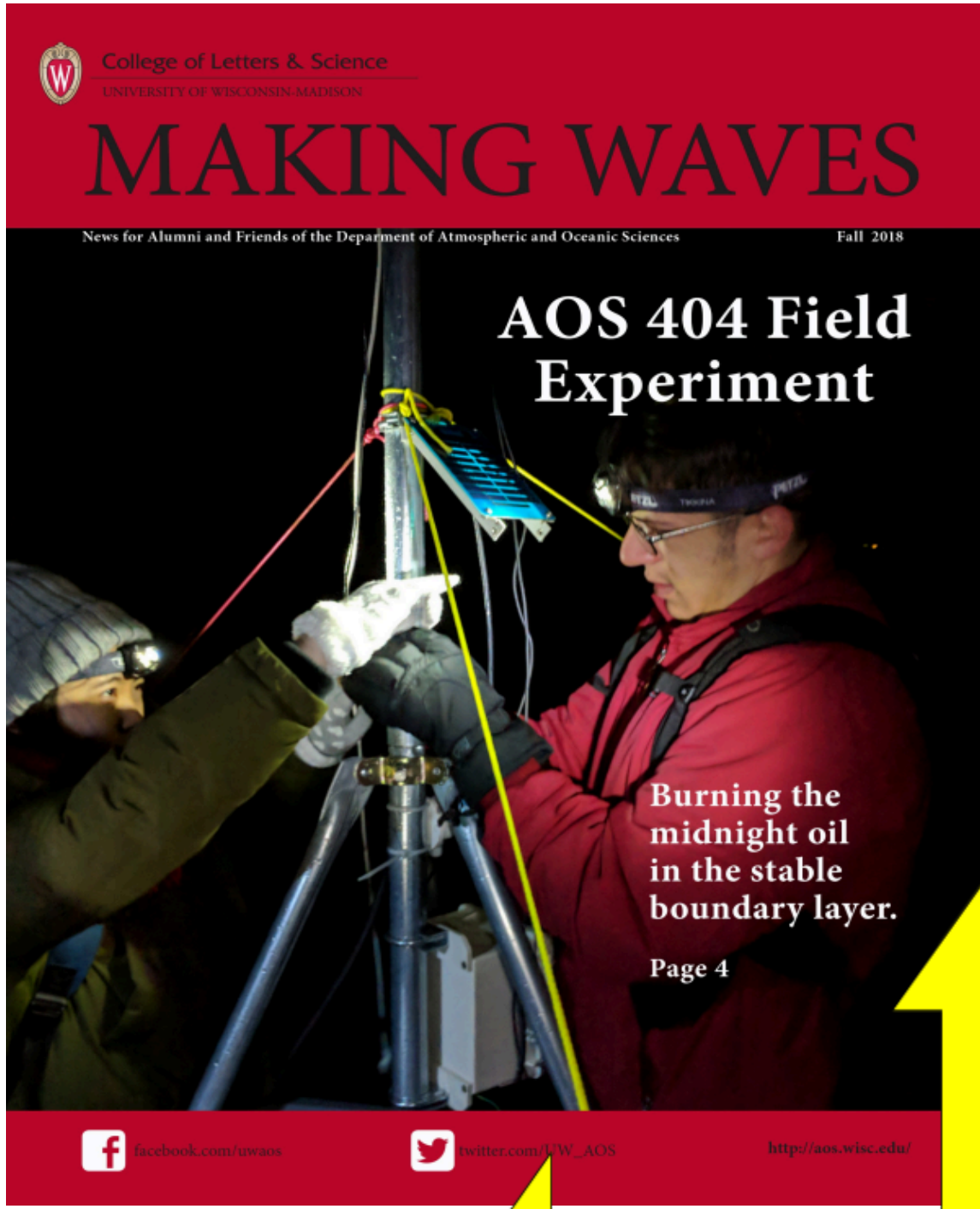
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# PUBLISHED FIELD RESEARCH



That's me with my friend Lena Tao (left)  
on the newsletter front cover!



## Save the date

The best social gathering at the annual American Meteorological Society meeting is scheduled for

**Tuesday, January 8th  
from 6-9pm,  
in room Encanto B  
of the Sheraton Grand Hotel in  
Phoenix, Arizona.**

As always, there will be great food and great conversation as we share memories and catch up on the news in each other's lives. We hope to see you there!

PHOTO BY JEFF MILLER

## CHAIR'S LETTER



As we progress down the road of renewal we are excited to inform you of a few of the many changes

occurring in the Department and University as I write these words. The University has pushed forward on its commitment to a new era of unprecedented growth, by enthusiastically supporting at every turn our requests to hire new faculty and develop new programs. As you may have noticed, over the past year, we have made accepted offers to two new faculty, Dr. Stephanie Henderson (tropical-extratropical connections, beginning Fall, 2019) and Dr. Elizabeth Maroon (ocean-atmosphere processes, beginning Winter, 2020). Dr. Luke Madaus of Jupiter Inc, will also open up an office in AOS where he will work with students wanting to intern in the area of big data analysis. We are continuing our search this year for at least one additional faculty hire, for which we have received an impressive group of applications. We are planning to modernize the 8th floor class rooms over the next 1-2 years while we hold our place in the second group of Departments slated for an eventual new building, perhaps as soon as the next decade. We also are working hard to secure at least one of several proposed cluster hires of an additional faculty, using a new University program that encourages cross-Department collaboration of this type.

We have also begun the process of creating a Professional Masters program, under the leadership of Professor Ankur Desai and with valuable guidance from the AOS Alumni board. If we can attain approval for our planned

implementation of the PM this Spring as we expect, we may begin reviewing student applications by this coming Fall for the 2020-2021 academic year. This exciting new program will create a 1-year applied Masters program, and be coupled to our Undergraduate degree in the form of a 5 year BS-MS program. We will split the current MS program into 2 named options, (1) the Professional Masters and (2) the Thesis Masters, (the traditional PhD track.). The PM will be based on a core sequence, required AOS classes and contributing classes from other departments (such as Business or Social Science) outside AOS. Jobs preparation classes and a summer internship will also be required. Undergraduates electing to pursue a PM (5-year Masters) would enter this program early by taking approved tracks of specialization in preparation for the PM. Market studies suggest that the PM could increase the AOS student numbers by as much as 50% or more within 4 years, sparking more growth of the AOS faculty and staff.

These are a few of the many ways our Department is growing under the new financial organization of Letters and Science and the commitment of the UW to a 21st century University.

On, Wisconsin!

Gregory Tripoli, Professor and Chair

Design: Sarah Morton, College of  
Letters & Science  
ON THE COVER Middle of the night  
DustTrak deployment. Julia Shates.

## AOS students lead popular atmospheric science class for the Wisconsin Center for Academically Talented Youth

by Andrew Dzambo, Brian Zimmerman, Luke Odell, Melissa Breeden, Zoe Brooke Zibton, Emma Sinclair

For the last three summers, graduate and undergraduate students from the University of Wisconsin's Department of Atmospheric & Oceanic Sciences have led an Atmospheric Science class through the Wisconsin Center for Academically Talented Youth (WCATY) program. Since the inception of the class, this course has been ranked among the most enjoyable and challenging classes offered by WCATY's Advanced Learning Program (ALP).

The ALP is one of several summer camps offered through WCATY, with this particular camp geared toward advanced high school students. Brian Zimmerman, a 2017 AOS M.S. degree graduate, formed the class during the summer of 2016 after teaching a 'Science of Sustainability' class for ALP in 2015 with the idea that a math- and physics-based weather and climate class would be challenging yet enjoyable for a group of gifted students. "I asked if they (WCATY) would let me teach a class

in weather and satellite meteorology, and they said to submit a sample syllabus," Brian recalls. The class was approved shortly after. "The class went over really well – with the help of the department (providing a classroom in the building) and Pete Pokrandt (for setting up computer accounts for the kids)." During the 2016 class, Brian took his students to the SSEC data center, the rooftop (to see the various instruments), and utilized Professor Ankur Desai's instrument lab and McIDAS as further instructional tools – tools that other WCATY classes simply don't have comparable access to. Along with Brian's class, Dr. Melissa Breeden (2017 instructor), Andrew Dzambo and Luke Odell (2018 co-instructors & current Ph.D. students) have taken students on a variety of field trips and incorporated guest lecturers from the AOS department, SSEC, CIMSS and Antarctic Research Center.

*Continued on page 7*



Group photo of the WCATY ALP 2018 instructional staff and students. Photo by Pete Pokrandt

## Alumni Engagement Board

This past October, the AOS Department opened its doors to alumni for the first annual 'Open House.' This was an initiative of the Alumni Engagement Board to encourage networking between alumni, current students, and former professors. Alumni were invited to spend the Friday of Homecoming Week catching up with friends, sitting in on classes, roaming the halls, and of course, dropping by the Friday afternoon weather watch. Many local alumni and a handful of out-of-towners shared stories over morning coffee and got to hear from Dr. Greg Tripoli on the latest happenings within the Department. Following weather watch, and after all of the Ian's Pizza was finished, a couple dozen undergraduate students had a question and answer session with alumni from various job sectors. Students asked several questions to the alumni as they offered career advice. As the day concluded, many attendees spread throughout campus enjoying the fall weather and Homecoming activities, such as the parade down State Street. The Alumni Engagement Board hopes to build on the event next Homecoming, so stay tuned for future invites.

- Co-Chairs: Dan Hartung (BS '07; MS '09) and Dr. Matt Sitkowski (PhD '12)
- Members: Dr. Tim Wagner (MS '06; PhD '11), Alex Kubicek (MS '13), Brian Miretzky (BS '06; MS '09), Tom Skilling

## Burning the midnight oil in the stable boundary layer

*As told by the UW AOS 404 Meteorological Measurements Fall 2018 students: Libby Carson, Andrew Kieckhefer, Vijit Maithel, Leo Mikula, Jacob Pozezinski, Claire Rubbelke, Monica Samsin, Julia Shates, and Madankui Tao.*

Under the light of the moon, stars and headlamps, a group of AOS undergraduate majors and graduate students trekked around a shallow gully and a gentle sloping cut-down cornfield to measure carbon dioxide, prepare aerosol monitoring instruments, and at sunrise, launch low-altitude weather balloons. The intense sleep-deprivation was all in the name of science as this fall semester's AOS404 Meteorological Measurements Class field experiment! The class had an opportunity to participate in the National Science Foundation (NSF) funded Stable Atmospheric Variability And Transport (SAVANT) field campaign in Mahomet, Illinois. The nine students, under guidance of instructors Prof. Ankur Desai, Prof. Grant Petty, and teaching assistant Ammara Talib, with support from NSF, the department, and alumni donations, spent a week in the field deploying their own instruments to study the nocturnal boundary layer and in particular, how it dissipates at sunrise and transitions to convection. The AOS group gained experience in planning and running experiments in the field all while having the opportunity to see how an intensive field campaign is run by the SAVANT Principal Investigators April Hiscox at University of South Carolina, David Kristovich of the Illinois Water Survey, and Junming Wang at University of Illinois along with their students and post-docs and staff from the National Center for Atmospheric Research (NCAR) Earth Observing Lab. During SAVANT, on calm stable nights, gully flow was

simulated with neutrally-buoyant smoke releases. The SAVANT team then monitored the atmospheric response by a series of towers with turbulence instrumentation, numerous atmospheric profiling instruments, and several scanning aerosol LIDARs.

The AOS student wrote a supplemental experimental design and bought their own instruments down to complement SAVANT's main experiment, including aerosol monitors, meteorological tripods, weather balloons, portable gas sensors, and a new temperature mapping drone. Below are some reflections of various instrumentation teams.

### Watching aerosols flow

Our aerosol team (Fig. 1), composed of two undergraduate students, Madankui Tao (left) and Andrew Kieckhefer (right), woke up at 4 AM each day of the trip to collect fine particulate matter (PM<sub>2.5</sub>) measurements before, during, and after sunrise using an optical instrument called a Dusttrak (TSI, inc.). Both students are undergraduate researcher studying air quality.

One DustTrak was staged inside of a gully while the other rested on top of a hill. They collected PM<sub>2.5</sub> measurements at these two sites in order to compare the impacts of sunrise on air quality in regions with shallow topography.

### Chasing balloons around farms

To capture boundary-layer transition, undergraduate AOS students Libby Carson, Leo Mikula and Jacob Pozezinski

launched small weather balloons designed for high-resolution vertical profiles of the boundary layer called Windsonds (Sparv Embedded AB). These were launched every half an hour, beginning at 6:30 AM and concluding at 8:30 AM. The instrument measured temperature, wind speed and direction, dewpoint and relative humidity. The group aimed to see how the lower atmospheric conditions changed throughout the morning as the sun began to warm the surface. The instruments, housed in a Styrofoam cup, can be dropped from the balloon by hot-wire while it telemeters GPS coordinates. A mad chase then ensues to find these cups spread out across central Illinois. Though a few were lost during the trip (one travelled over 18 miles), the group got some wonderful data and look forward to sharing the results.



MORNING WINDSOND LAUNCH

PHOTO: JACOB POZEZINSKI

### Ghostbuster or Gas Scouter?

Claire Rubbelke and Monica Samsin, accompanied by the class TA Ammara Talib, used two devices to collect carbon dioxide (CO<sub>2</sub>) concentrations in the gully: a CO<sub>2</sub> handheld probe (CO<sub>2</sub>meter, Inc.) and a high-end laser based portable sensor, the GasScouter (Picarro, Inc.). Overnight, the students attached the CO<sub>2</sub> probe onto a weather

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# RESEARCH WITH STAKEHOLDERS

## **Joining the SAVANT Campaign: Exploring Stable Boundary Layers**

AOS 404: Meteorological Measurements

Libby Carso, Andrew Kieckhefer, Leo Mikula, Vijit Maithel, Jacob Pozezinski, Claire Rubbelke,  
Monica Samsin, Madankui Tao, Julia Shates

### **1. Abstract**

During the Fall of 2018, the Measurements class in the Atmospheric and Oceanic Sciences (AOS) department from the University of Wisconsin - Madison joined the SAVANT (Stable Atmospheric Variability ANd Transport) team in Mohamet, IL to observe the ongoing field experiments and conduct their own measurements. Over the four day trip, students were able to observe and measure data relating to fine particle distribution, solar energy balance and general atmospheric characteristics using a variety of instruments ranging from windsonds, to dusttracks, to CO<sub>2</sub> gas scouters and setting up their own weather stations. This data allowed the students to explore properties of the atmosphere in a stable, nocturnal boundary layer and how the boundary layer slowly evolved and eroded following sunrise.

### **2. Introduction**

#### **2.1. Literature Review**

In general, we are keeping these questions in mind while developing this proposal and the background to our research questions: How we can supplement the SAVANT field campaign with our work? And how are our research questions distinct? As we continue working on the project, we are working together to compile a growing list of sources to develop our understanding of boundary layer stability, radiation, particle dispersal and ecosystem response.

While the SAVANT campaign explores converging flows in the stable nocturnal boundary layer, our proposal supplements the study by expanding the scope of the work to include the sunrise transition. By also investigating the carbon dioxide (CO<sub>2</sub>) fluxes, we will provide insight into how the ecosystem responds to cold pooling in shallow topography.

The characteristics of the stable boundary layer in Champaign-Urbana, Illinois have interested scientists for many years because of the relatively flat surface. In the late 1990s, there was a field campaign (Flatland96 Experiment) which had a wide variety of instruments over the agricultural terrain and through plots of soy and maize measuring surface fluxes, radar reflectivity and winds using sonic anemometers. The experiment also had 2m towers for temperature and humidity sensors. Avengine et al (2001) also analyze the growth of the CBL at a station in the Netherlands (Cabauw, station KNMI) with similar terrain, measurements and a 200m instrumented tower to study the boundary layer. There they examined the standard deviation of temperature, wind speed and direction in combination with profiler reflectivity to determine the onset of the CBL.

Beare (2008) use a large-eddy simulation (LES) modeling study to define the morning transition of the boundary layer. This work focuses on the role of shear in the SBL for the transition between the stable boundary layer (SBL) and the CBL. For this work, they define the transition between the SBL and CBL to occur when the sensible heat flux changes sign (from negative during the night to positive). By using the LES, they are able to determine the

sensitivity to shear during the morning transition by testing different geostrophic wind scenarios. They find that the transition is dominated by shear and the CBL, in contrast, was primarily buoyancy driven.

By exploring the stability, shear and radiative and turbulent fluxes during the SAVANT field campaign, we hope to be able to also comment on the role of shallow topography in the dissipation of the stable boundary layer. There is a connection between air pollution and dispersal of particles because the timing of the transition between the nocturnal layer and convective boundary layer (CBL) occurs during peak emissions during morning rush hour (Angevine et al, 2001). Understanding how shallow topography influences the transition may be relevant for populated areas that are situated on topography.

Mahrt, et al. 2001 and Soler, et al. 2002 both focus on the analysis of a weak gully flow that formed over the same shallow gully during CASES99. The gully consisted of a small valley with a gradient of 5%. 2D sonic anemometers and thermistors were placed across the gully to capture this flow. It was observed that a weak, thin flow occurred in the early evening on nights that had clear skies, minimum winds, and maximum temperature stratification. This flow lasted until the middle of the night, when turbulence-driven mixing and wind intensification disrupted the stratification. After midnight, a period of warming was observed, which the authors associated with downward momentum transport and minimal gully flow. In the early morning, decreases in downward mixing and turbulence prompted near-surface temperature decreases and gully flow was reestablished as the air re-stabilized. At the gully bottom, the temperature reached its minimum for the night right after sunrise.

A secondary experiment was implemented to visualize transient and spatial variability. Smoke releases were tracked in an effort to illustrate how “transient mesoscale motions lead to significant meandering”. Mahrt, et al. continued with data analysis and calculation of surface heat flux and characterization of the surface energy budget. Soler et al. explore the calculations regarding turbulence events that compromised the gully flow.

Pypker et al. 2007 investigated the potential for using nocturnal cold air drainage flow to monitor ecological processes. This investigation took place in a forested valley/watershed. Two aspects were explored: carbon isotope composition of the nocturnal ecosystem respiration (for the purposes of our experiment, we won't examine their methodology) and estimates of the CO<sub>2</sub> flux in the watershed that is advected in the drainage flow. Observation towers were assembled and deployed with thermistors, mechanical and sonic anemometers, and radiometers. A definition of cold air drainage was established, with flow needing to meet three criteria: (1) wind direction must be down valley, (2) wind-speed must have a local maximum near the ground, and (3) the net radiation measured above the tower must be negative, indicating the nocturnal surface cooling.

Results of the field measurements showed that CO<sub>2</sub> concentrations varied vertically with height, as well as with time. Concentrations during the day reached their highest concentration near the ground. As the air began mixing throughout the night, CO<sub>2</sub> concentrations from all heights converged (at this time the potential temperature profile became nearly isothermal). Following this mixing episode, concentrations rise steadily until sunrise, when drainage



remained well mixed, but concentrations began to decline. When the air again became stable, the vertical gradient of CO<sub>2</sub> concentration was re-established.

Basic calculations were made to provide an estimate of respiration that occurred within the watershed. Final calculated values of flux showed a range from 0.3 to 1.3 mol/s between 20:00 and 05:00. The authors discussed error sources for their calculations, as their flux values seemed low, ultimately suggesting that the remainder of the respired CO<sub>2</sub> may have been advected above or out of range of the measurement towers.

The concentration of PM<sub>2.5</sub> is affected by solar-driven boundary layer growth and decay: while short-lifetime gases reach minima at sunrise and maxima in the evening, fine particulate matter does not show a large diurnal variation in previous studies (Tanner et al., 2005).

## ***2.2. Hypothesis/Questions***

### ***2.2.a. Weather Station***

Are weather conditions different inside and outside the gully, and how does gully flow affect these conditions?

### ***2.2.b. CO<sub>2</sub> Tracker***

Gully flow will occur if conditions are ideal. However, flow will be thin and easily affected by external disturbances but can be observed through a rapid shift in CO<sub>2</sub> concentrations. Due to stratification, flow would decrease CO<sub>2</sub> flux to the boundary layer due to stratification.

### ***2.2.c. Windsong***

The goal of the Windsong Team was to observe how the stable boundary layer in the atmosphere eroded during sunrise. Their observations would be obtained through a series of instrument launches throughout the morning. With each launch, the windsong instrument takes measures of air temperature, moisture and wind direction. We hypothesize that there will be a distinct shift in the temperature profile. Changes in wind speed and direction may be good indicators of a changing boundary layer. We expect the boundary layer to be heavily stratified at night and more unstable during the day. Cloudy conditions will delay the boundary layer destabilization. Clear conditions will provide the optimal observations as the sun will be able to directly warm the surface and we will be able to monitor resulting changes throughout the layer in real time.

### ***2.2.d. Surface Energy Balance***

We hypothesize that the surface energy balance at sunrise (or shortly after) will have a clear shift during the transition between a stable and unstable boundary layer. Additionally, we hypothesize there will be an indication of gully flow in the SEB.

### ***2.2.e. DustTrak***

With the change in boundary layer conditions before and after sunrise, we are curious about how it will influence the concentration of particulate matter present during the transitional

periods. We hypothesized that PM<sub>2.5</sub> concentration would increase after the sunrise at a higher rate inside the valley versus outside because of the outside surface heats up faster.

### **3. Methods**

#### ***3.1. Weather Stations***

Two HOBO weather stations, Station B and Station E, were setup at locations outside and inside the gully respectively (Figure 3.4.1). There is a height difference of 13.5m between the two locations, with Station B being at the higher elevation. Each weather station was equipped with a cup anemometer, a wind vane, a temperature and humidity sensor, a pressure sensor, a solar radiation sensor, and a data logger with battery. Temperature, relative humidity, wind speed and direction, incoming solar radiation, pressure, gust speed, and dew point temperatures are recorded by the weather stations at a one-minute resolution from 14 October afternoon to 17 October morning. The data was analysed with the help of HOBOWare software and helped compare the weather conditions inside and outside the gully.

#### ***3.2. CO<sub>2</sub>***

A handheld CO<sub>2</sub> probe and a gas scouter were used to measure CO<sub>2</sub> concentrations. On one night, October 15, the handheld CO<sub>2</sub> probe was attached to a weather station at the base of the gully. Measurements were recorded for two time periods and analyzed to determine if CO<sub>2</sub> was advected through the gully to ultimately pool in the base of the gully. On October 16, the gas scouter was taken through the gully to collect concentrations during three time periods, each starting at 12 AM, 1 AM, and 2 AM local time. By taking CO<sub>2</sub> measurements throughout the gully at various times through the night, we were able to observe the advection of CO<sub>2</sub> and the evolution of air flow as the night progressed.

#### ***3.3. Windsound***

There were a number of processes that needed to be completed each launch to ensure that we were able to collect accurate data that would help us determine if the boundary layer destabilizes after sunrise. On launch mornings, the team would arrive at the field site no later than 6:15 A.M. This would ensure that we were able to set up the launching station and correctly link all necessary software on the computer with the windsounds. When we arrived at the field, the group set up our table and all the necessary equipment. This would include, the windsound instruments, the field computer and the helium canisters. We would then log into the computer and initiate the windsound software. All data would be collected through this program. The windsound would then be activated to establish a radio and GPS link between the computer and the instrument. It was imperative that this link is established as it is the sole pathway for collected data to be sent to the computer. Once this connection was established, a small scale weather balloon was filled with helium and tied to the windsound. A cut-off level of 400 m above the surface was established in order to get an accurate profile of the boundary layer. An electrical current would be sent through a cable in the windsound in order to cut the string attached to the balloon so that the sonde would fall back to the surface. When all steps had been completed, we would launch the balloon and watch real-time data be sent back to the computer. The windsound



would take approximately 10 minutes to reach the desired cut-off height. A GPS coordinate would be sent to the computer when the windsound had reached the ground. The retrieval team would then travel to the location to find the instrument.

### 3.4. Surface Energy Balance

To begin to assess the SEB at sunrise and to further explore the stability of the atmosphere, we use radiative and turbulent fluxes measured by radiometers from SAVANT NCAR observation towers. Specifically, we calculate the SEB with net longwave (**LW**), net shortwave (**SW**) broadband fluxes for the Net Radiation (**R<sub>n</sub>**), and latent (**LH**), sensible (**H**), soil (**G**) heat flux, and assume the existence of a residual and error.

$$R_n = SW(in-out) + LW(in-out) = H + LH - G + Residual \text{ (\& Error)}$$

$$LH = \underline{w'h2o'} * Lv \quad Lv = \text{latent heat of vaporization}$$



Figure 3.4.1: Schematic showing the placement of various instruments at the field site

$$H = \underline{w'T'} * \rho * Cp \quad \rho = P/(Rd*Tv) ; Cp = \text{specific heat at constant pressure}$$

The data we use are pre-quality controlled and were provided by Steve Oncley at the Earth Observation Laboratory (EOL) and is 5-minute time resolution. We apply a 30-minute running mean to the data, which smooths the terms and helps resolve turbulent eddies that develop over longer time scales ( $\geq 30$  minutes).

For latent heat flux (**LH**), we use the covariance (2nd moment) of the anomalies of vertical motion ( $w'$ ) and water vapor ( $h2o'$ ). For sensible heat flux (**H**), we use the covariance for of the anomalies of vertical motion ( $w'$ ) and temperature ( $T'$ ). For the soil heat flux (**G**), we changed the sign in the surface energy balance equation from Bonan (2008) to keep this variable consistent with the direction of the other variables in the equation.

We use observations made from the release tower and the lower convergence zone (B) at 20m height. The LW fluxes were measured by thermopiles; we calculate the density of air( $\rho$ )using the pressure observations and the virtual temperatures calculated from sonic anemometers. The soil heat flux values are measured with a soil heat plate 5 cm below surface.

To identify the impact of shallow topography on the flow of heat/energy, we calculate the residual from the SEB equation. Then, we compare the residual values through time at the release tower and the convergence zone.

### 3.5. DustTrak

After the weather stations are setup and operating, begin DustTrak assembly. The two devices will be secured to the weather stations; therefore, weather stations must be set up before DustTraks. Begin with preparing the waterproof casings. First, confirm that battery level is at 99% or higher before starting. Screw the PM<sub>2.5</sub> filter onto the intake hole. Program DustTrak with desired DustTrak parameters. In this case, collecting PM<sub>2.5</sub> concentrations every second.

To prepare the waterproof casing, puncture a hole the side of 1m tubes. Hole should be just larger than the diameter of the 1m tubing. Secure one end of the 1m tubing 1.5 meters above the ground. The tube should open to the sky and the rest should feed below 1.5 meters. Feed the other ends of the tubes into cases about  $\frac{1}{3}$  the length. This will make the next step easier. Insert the end of PM<sub>2.5</sub> filter into the end of the tube inside the casing. Once snug, pull out any tube excess so the DustTraks can rest comfortably in the cases after inserted. Place the cap on waterproof cases and close on all sides.

Once the DustTraks are inside the secured cases, we configured two Zip Ties to each side of the containers and tied them around the weather stations' legs. While this is going on, we hear the DustTraks buzz, meaning they are working properly. Now we wait for our data to collect. In the meantime, we supported and observed other portions of the field experiments. Let DustTraks collect data for 3 hours. Overall, 6 sets of data should be collected, each with 3-hour intervals of PM<sub>2.5</sub> concentration readings. Devices should be started 1.5 hours before sunrise and then turned off 1.5 hours after sunrise. After data collection and retrieval, we need to have the field data logged on the field laptop. Connect the two devices and upload the data then save. Make sure DustTraks have enough battery life to support another morning of data collection.

## 4. Results

### 4.1. Weather Stations

Different weather conditions were observed during the three nights of experimentation. The night of 14th October was rainy along with high winds associated with a cold front passing through the area. Night of 15th October was clear with very low winds, making it ideal for gully flow. An IOP was conducted by the SAVANT team on this night. Night of 16th October, although clear, had relatively higher wind speeds. Data from the two weather stations is shown in the following figures, with data from Station B

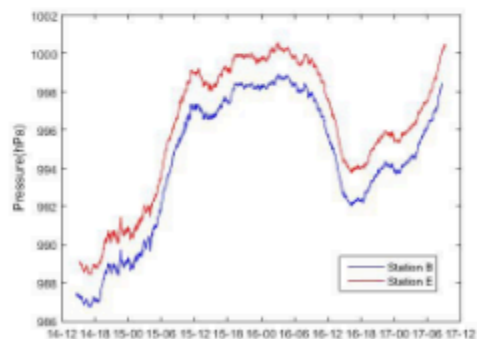


Figure 4.1.1: Pressure measured in hPa at Station B (blue) and station E (red).

(outside the gully) in blue and data from Station E (inside the gully) in red.

From hydrostatic balance, a pressure difference of around 1.6hPa is expected between the two locations due to the 13.5m difference in heights, with pressure being higher at lower location (Station E). This is observed in figure 4.1.1.

The wind vane is setup such that values closer to  $0^\circ$  (or  $360^\circ$ ) represent winds coming from the north,  $90^\circ$  represents winds from the east,  $180^\circ$  represents winds from south, and  $270^\circ$  represents winds coming from the west. Figure 4.1.2 shows wind speed (top) and direction (bottom). Strong winds are coming from a north, north-west direction on 14th night and 15th morning, which can be expected during passage of a cold front. Wind flows from a western direction in general for the rest of the days. Winds were considerably calmer on the night of 15th October. Also, winds inside the gully (Station E) at Station B (blue) and station E (red) are clearly weaker compared to winds outside the gully.

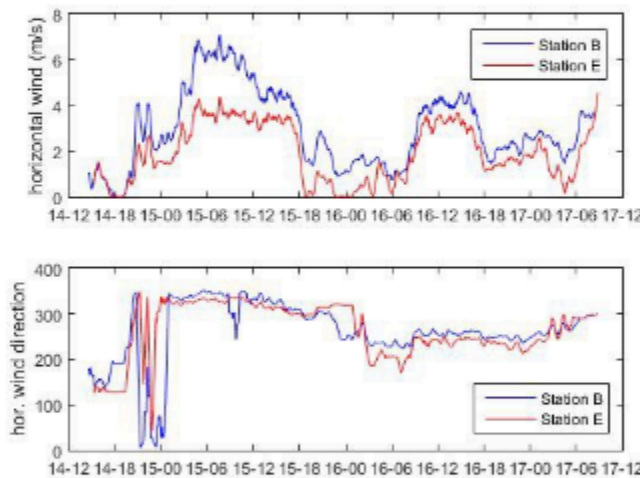


Figure 4.1.2: Wind speed (top panel) and wind direction (bottom panel) at the two weather stations

Figure 4.1.3 shows the temperature measured at the two weather stations. Diurnal cycle is clearly visible in the time series. Temperature measured at the lower convergence tower is also plotted in black to compare with another location inside the gully. Overlap of red and black curves, show that temperature is fairly uniform along the gully at all times. Furthermore, the temperature is almost the same inside and outside the gully for most times during the observation period, with the exception on the night of 15th October, when temperatures inside the gully are cooler compared to outside.

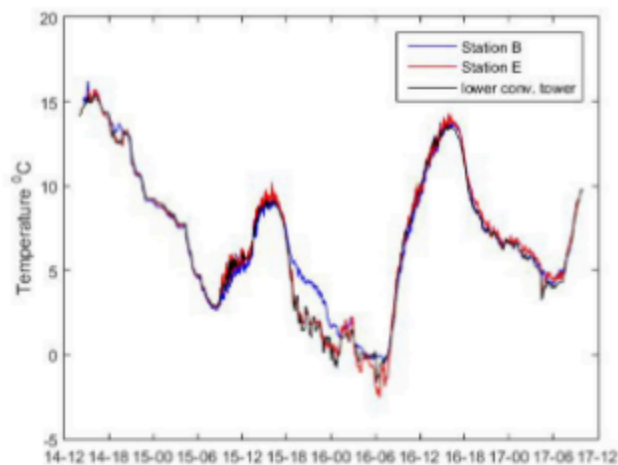


Figure 4.1.3: Temperature at different locations. Station B (blue), Station E (red), and lower convergence tower (black)

#### 4.2. CO<sub>2</sub>

The conditions at midnight on October 15 were 48°C with occasional gust speeds of 3.94 mph from the north and northwest; there were precipitation and overcast clouds. The conditions at midnight on October 16 were 32.9°C with little-to-no winds from the west and northwest; there were no precipitation or extensive cloud cover. On October 17, the temperature was 44°C with a wind speed of 5 mph from the southwest direction.

Data from the gas probe varied little through the period of collection. The peak value was 391 ppm, and the low was 383 ppm. Overall, the concentration remained fairly constant through the time of observation.

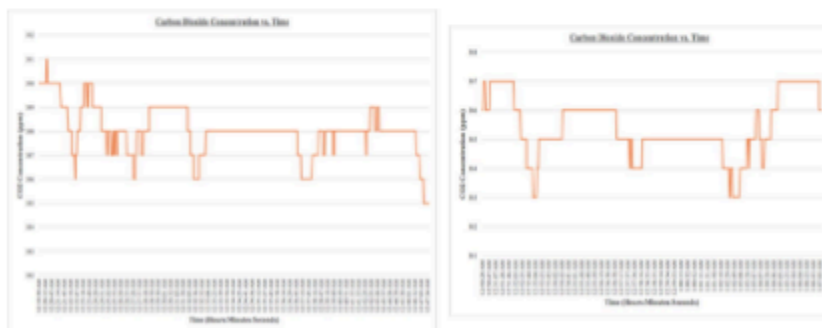


Figure 4.2.1: Carbon di-oxide concentration measured from probe inside the gully on 15th October.

Data from the gas scouter during all three runs showed lower concentrations of around 430 to 470 ppm at the top of the gully, with an increasing trend of concentrations toward the base of the gully. The highest concentration was 1052 ppm. Higher concentrations occurred later in the evening.

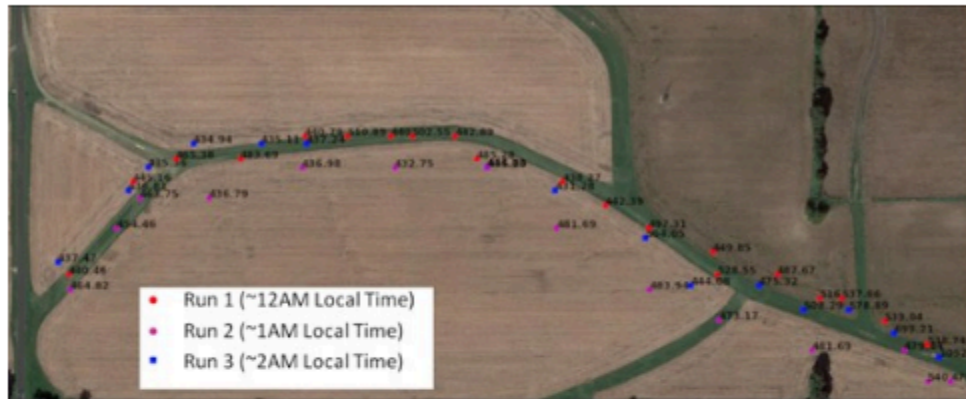


Figure 4.2.2: Image showing spatial map of carbon dioxide concentration as measured during the three gas scouter runs

#### 4.3. Windsound

The best data from the three days of measurements came from October 17th. This day provided the group with clear conditions and calm winds. We were able to launch the windsounds every 30 minutes and had no issues with any of those launches. The data we received from the instruments showed the slow destabilization of the boundary layer throughout the morning.

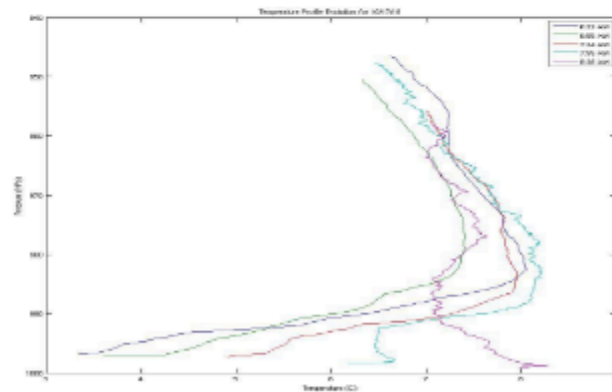


Figure 4.3.1: The temperature profile of the atmosphere during the sunrise transition is shown. Each colored plot corresponds to a morning launch (occurring every 30 minutes). Note the gradual warming near the surface, emphasizing an evolving boundary layer.



The figure above shows the temperature profile throughout the morning of October 17th. Each line is color coded to correspond to launch times of the windsounds. The early morning launches all occurred before sunrise (7:04 AM on this day). As shown, the temperature profile at the 6:22 AM and 6:55 AM launch illustrates a stable boundary layer just above the surface. Once the sun began heating the surface and mixing near the surface occurred, the temperature near the surface began to increase. The temperature profile from the last morning launch, happening at 8:36 AM, shows a more unstable layer near the surface as compared to the first windsound launch at 6:22 AM on the 17th. The lapse rate is measured to be near the dry adiabatic lapse rate (9.8 degrees C/Km).

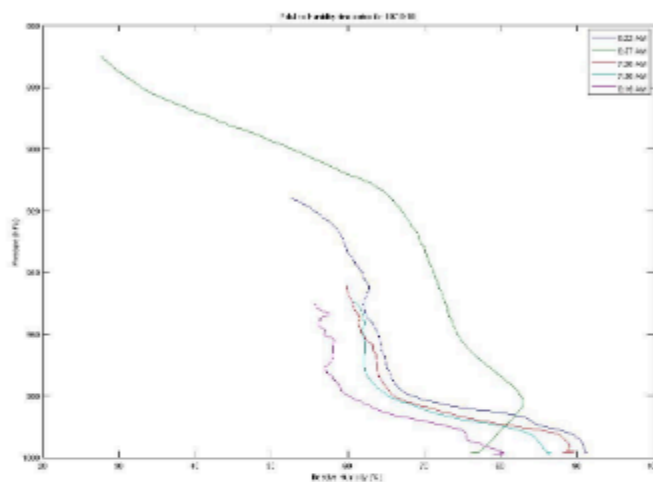


Figure 4.3.2: Plot of the measured atmospheric relative humidity before, during and after sunrise on October 16th. Different launch times are shown in varying colors.

There were a number of other interesting results from the data that was collected by the windsounds. During sunrise, the relative humidity observed slowly decreased once the sun rose. This is not a surprising result as the temperature increased while the dew point remained constant. The wind speed just above the surface (100-200 meters) gradually increased as well. This resulted in higher atmospheric shear at lower levels, enhancing mixing near the surface.

#### 4.4. Surface Energy Balance

For the results of the SEB analysis, we focus on Monday/Tuesday 15-16 October, which was one of the nights that the SAVANT researchers forecasted conditions favorable for a stable, nocturnal boundary layer. In the figures below, we can identify a shift in all of the radiative and turbulent fluxes at sunrise (~7:05am). Net radiation switches sign to positive with increasing shortwave radiation from the sun- the surface is no longer radiatively cooling. It is apparent that there is a time lag from when Rnet switches sign and when the turbulent fluxes and soil heat flux respond and switch sign. With the exception of the soil heat flux G, H takes the longest time to respond to the incoming shortwave radiation. In contrast to the fluxes, the SEB residual appears to lead (or aligns very closely with) the Rnet when it begins to increase.

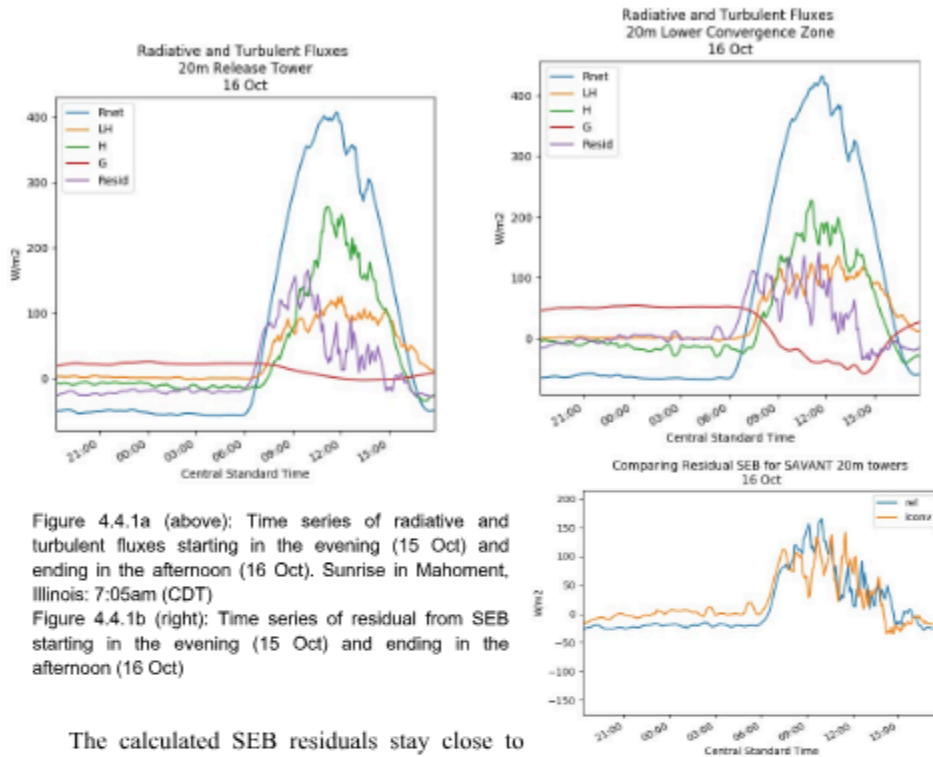


Figure 4.4.1a (above): Time series of radiative and turbulent fluxes starting in the evening (15 Oct) and ending in the afternoon (16 Oct). Sunrise in Mahomet, Illinois: 7:05am (CDT)

Figure 4.4.1b (right): Time series of residual from SEB starting in the evening (15 Oct) and ending in the afternoon (16 Oct)

The calculated SEB residuals stay close to zero throughout the night at the release tower and the lower convergence zone. During the nighttime hours, however, the lower convergence appears to have more blips in the time series. The two residuals have a correlation of 0.8. For the lagged correlation at 5 minutes, the value decreases to 0.79 and continues to decrease to 0.76 at 20 minutes.

#### 4.5. DustTrak

We have first compared the  $PM_{2.5}$  concentrations between two weather stations. We hypothesized that the concentration for  $PM_{2.5}$  is higher outside of the gully since it heats up more quickly. Through a T-test between the two stations, we found strong support to our hypothesis with P-value smaller than 0.01, which indicates that the concentration of fine particulate matter is higher on the Upper Tower outside of the valley. Then we did a similar test by dividing data to before and after sunrise. Due to an increase in temperature and disturbance in the stable boundary layer after sunrise, we hypothesized that  $PM_{2.5}$  concentration would also increase. The statistical test gives a p-value smaller than 0.01 and provides strong support to our hypothesis. In sum, we found that  $PM_{2.5}$  concentration would increase after the sunrise at a higher rate inside the valley versus outside.

Besides comparing  $PM_{2.5}$  for different locations and period, we try to associate it with meteorological variables measured by the weather stations. Through Pearson correlation analysis, we found that temperature is positively related to  $PM_{2.5}$  concentration (with Pearson correlation coefficient of 0.57) while relative humidity is negatively associated with PM concentration (with Pearson correlation coefficient of -0.62). Solar radiation and wind speed and direction do not show a linear relationship with PM.

Of the weather variables we compared with  $PM_{2.5}$  concentrations, relative humidity and temperature were the most significant.

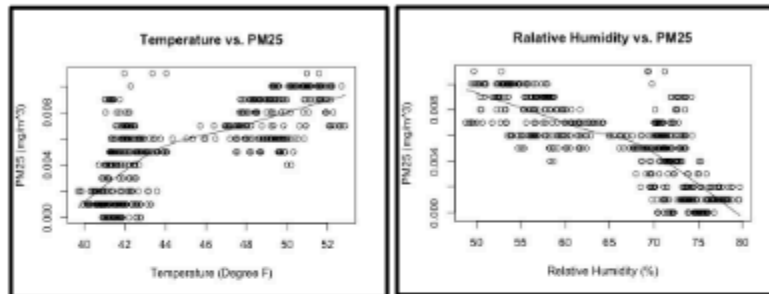


Figure 4.5.1:  $PM_{2.5}$  concentration as a function of temperature (left) and relative humidity (right)

The above figure shows relationships between  $PM_{2.5}$  and temperature (left) and  $PM_{2.5}$  and relative humidity (right). On the left-hand side, the data indicate that an increase in air temperature is related to higher concentrations of  $PM_{2.5}$ . We observe the opposite situation regarding relative humidity on the right-hand side — less relative humidity in the air related to a decrease in the observed  $PM_{2.5}$  concentrations. These results are consistent with what we would expect.

## 5. Discussion

### 5.1. Weather Stations

Cold air drainage flows in shallow topography are very sensitive to the local weather conditions. Clear skies, and low wind conditions are most ideal for such a flow to be established (Mahrt, et al. 2001). Such a flow of cold, high density air through the gully should be reflected in temperature measurements inside the gully. Figure 4.1.3 shows that on the night of 15th October, when there were low wind and clear sky conditions, temperature inside the gully is lower. On the other two nights, when the weather conditions did not allow for gully flow to be setup and there is no cold air flowing through the gully, no temperature difference is observed between locations inside and outside the gully.

### 5.2. $CO_2$

Data from both the probe and the gas scouter supports the idea that katabatic gully flow forms in environments where the air is densely stratified. This occurs on nights when there is low cloud cover, which allows for more radiative cooling and the formation of a nocturnal boundary



layer (Soler, et al., 2002; LeMone, et al., 2003). Observations from the gas probe showed that on a night with overcast skies, there was not enough radiative cooling to induce the stratification necessary for the formation of the boundary layer.

The gas scouter recorded thousands of concentrations, and not all could be plotted. The investigators used a formula to parse through the data and create a separate data file which contained every 150<sup>th</sup> data point. This was done to create a clear and easy to read map of the spatial distribution of CO<sub>2</sub> concentrations through the gully.

### **5.3 Windsound**

There were a number of challenges that arose during our data collection that required special attention or alteration of launch times. These challenges were encountered early and we were able to accurately address the issues. On our first official launch day, Monday, October 15<sup>th</sup>, the weather proved to be anything but ideal. A thick lower cloud layer prevented any substantial heating by the sun, significantly affecting the measurements taken by the windsound. However, this was not the biggest issue we encounter due to weather. This day also was the windiest of the week, causing major issues for the single windsound we launched in the morning. As we were still learning the proper settings for each launch, the cut-off point for the sond was set at (number), much higher than the final level that was determined for the group. The strong wind, combined with a high cut-off level, allowed the windsound to travel very far from the launch location. This result encouraged the group to limit the number of launches initiated on windy days. If the wind was too strong, launches would be postponed for the day.

These issues were fairly minor and only required slight alterations in scheduling for launches and settings for each windsound. Our greatest challenge, however, was directly related to hardware of the windsondes. As previously mentioned, we were required to set a cut-off level for each windsound prior to the launch. Once the windsound passed this level in the atmosphere, an internal wire would heat up and burn the string connecting the windsound to the weather balloon. The windsound would then fall to the ground where it would be retrieved by a team of students. During one of the initial launches, this cutting mechanism failed and prevented the windsound from returning to the ground. Though we were still able to collect data, we were unable to retrieve the windsound. In the end, we were able to construct a beautiful atmospheric profile as the windsound rose far past the desired cut-off level. This specific windsound also traveled nearly 18 miles from its original location before we lost radio contact. In all, we lost a total of 4 windsondes, 2 from failed shutdown mechanisms and 2 due to inability to locate them on the ground (one falling in a dense forest and another landing in the middle of a lake).

### **5.4. Surface Energy Balance**

The sensible heat flux (H) changing sign is an indication of the destabilization of the stable, nocturnal boundary layer (Bonan, 2008). Turbulent eddies are able to develop within the boundary layer, but the measurements show that the transition is not immediate- there is a time lag from Rnet increasing and H increasing (and ultimately changing sign). The soil heat flux (G) switches sign from positive to negative as the ground is no longer a source of heating to the atmosphere but begins to absorb energy. The increasing LH is evidence for evaporation within

the boundary layer, and the curves for the release lower convergence towers appear coherent. This suggests that evaporation may not differ largely from the two towers, so, it appears to occur at a larger scale independent of the shallow topography.

The changes in the residual from SEB equation at both towers align very closely with the Rnet changes. This term in the equation may identify the start of the transition to a destabilized atmosphere. During the night time, the residual at the lower convergence tower is consistently higher than at the release tower. Considering the equations are very close to being balanced, and that turbulent eddies from H are close to zero, we may be able to infer gully flow from the release to the lower convergence zone. The correlation (0.8) between two locations is highest when looking at the same time step; a lag 5 - 20 minutes decrease the value, which contrasts with what I expected. It is possible that using a 5-minute running mean may remove some of the phase lag in the flow along the gully.

### **5.5. DustTrak**

By comparing the variation of DustTrak concentrations of  $PM_{2.5}$  before and after sunset, as well as inside and outside a gully, we found that  $PM_{2.5}$  concentration is significantly higher outside the gully after sunrise. This study supported both of our hypotheses. This finding confirms our expectation that disturbance of the nocturnal stable boundary layer after the sunrise will lead to an increase in  $PM_{2.5}$  level.

We used the weather station data to justify the pattern we observed for  $PM_{2.5}$ . We find a moderate positive correlation between PM concentration and temperature and a moderate negative relationship between PM concentration and relative humidity. This is reasonable since higher temperature, and the drier atmosphere is often related to higher PM concentration and explains what we observed for the morning transition. Besides, the temperature increased after sunrise, which disturbed the stable nocturnal boundary layer formed during the night. As stratified air started to move, observing an increase in PM level at 1.5 meters was unsurprising.

We also did the same linear analysis for wind speed and direction but did not find any relationship between PM concentration and any of these two variables with a small magnitude of Pearson correlation coefficients. We expected to see a connection with at least one variable for a few reasons. Fine particulate matter may result from the wind blowing in  $PM_{2.5}$  from an area of higher concentrations. Most of the  $PM_{2.5}$  arrived from southwesterly winds. While this was a trend, it was not significant. Wind speeds had little to no relation to the  $PM_{2.5}$  content in the air.

There are certain limitations for this study that need justification for future research. Firstly, we only collected  $PM_{2.5}$  data for three days with a three-hour duration each day. More data points are necessary to confirm the pattern indicated by this study. Secondly, one of the DustTraks automatically turned off as a result of the cold temperature before sunrise. Future measurements should pay attention to the operation range of the measurement tool to avoid such failure.

## **6. Conclusions**

### **6.4. Weather Stations**

The data collected from the two weather conditions helped study the weather conditions inside and outside the gully. The difference in pressure at the two locations matched with the expected pressure differences due to elevation from hydrostatic balance. Wind speeds were noticeably weaker inside the gully, possibly because of a shielding effect by the shallow topography. Gully flow when setup brings cold dense air into the gully, making the gully cooler than outside regions.

## **6.2. CO<sub>2</sub>**

Gas scouter data highly suggests the presence of a nocturnal boundary layer, which then manifested as down-slope flow. Run 1 implies a slight drainage present, with measurements from the top and base of the gully only varying by about 100 ppm. Run 3, started at ~2AM, suggested a strengthening of flow as the night progressed. At this point the concentration of CO<sub>2</sub> at the top of the gully was recorded at 437ppm and increased through the gully with a final reading of 1052 ppm at the base of the gully. Presence of a cold pool (Whiteman, 2000) at the base of the gully is thus inferred.

Run 2 recorded data from a track adjacent to the gully. This yielded reading of CO<sub>2</sub> concentration that showed little variance as the gully was traversed. The reading at the top of the gully was 432.74 ppm and the average CO<sub>2</sub> concentration at the base of the gully was 480 ppm. This data shows that a weak general flow may have formed outside of the gully, but more evidence would be needed to verify this claim.

The third hypothesis was unable to be tested because of time and money restrictions.

## **6.3. Windsond**

The group hypothesized that there would be a noticeable shift in the temperature profile of the atmosphere during the transition for a stable to unstable boundary layer during sunrise. After conducting numerous launches during our three-day collection period, the data received supports this hypothesis as there was a distinct change in the temperature near the surface throughout sunrise. The temperature gradually rose, causing the boundary layer to become more unstable after sunrise. Changes in wind speed and relative humidity also illustrate the atmospheric transition to a more unstable boundary layer.

## **6.4. Surface Energy Balance**

To better understand the transition from stable to convective boundary layer, it is important to assess the changes in radiative and turbulent fluxes for more than one night/morning. While the night & morning of 15-16 October was ideal for exploring the research questions, without other cases, it is uncertain whether the characteristics of the Surface Energy Balance in this paper represent most transitions or if the results were anomalous. Realistically, this is a reasonable task that can be pursued. There are two different approaches to studying the transition in the future with the NCAR/EOL tower data: 1) analyze all days that were flagged as having a stable nocturnal boundary layer to characterize the transition or 2) use criteria from the 16 Oct. case to identify these days and compare to the days flagged by SAVANT researchers.

In addition to more data, this section of the work can also be expanded by exploring the SEB terms through more statistical methods such as using the cross-correlation function to understand the lead/lag and relationships. In addition, significance testing is needed for correlation values.

#### **6.5. *DustTrak***

Our group hypothesized that  $PM_{2.5}$  concentration increases after the sunrise and is higher outside of the gully during this period. Based on data collected before and after sunrise, both hypotheses were supported with statistical significance. During the morning transition period, level rises as the nocturnal boundary layer are disturbed by the heating of the land surface, and this heating process happens at a faster rate outside of the gully. We also found moderate associations between  $PM_{2.5}$  with temperature and relative humidity. No direct association with wind speed and wind direction was found in this study.

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# RESEARCH SCHOLARSHIP

Summer Research Scholars Runzheimer's International Research Public Summary 2013

## Introduction

Drought plagues about 40% of America's soil according to NOAA. Much of that soil includes agricultural harvests that depend on water to grow. 83.4 million acre feet of water have been used on average since 2013 through 2018 according to USDA. According to the National Library of Medicine, "plants must make do with the water in their immediate surroundings, whether it be abundant or scarce."

## Research Question

Potassium is known to support moisture absorption according to the National Environmental Methods Index. If we had four separate environments to grow radishes in, would the crop yield differ based on the water and potassium content?

## Methods

|                                                                 |                                                                |
|-----------------------------------------------------------------|----------------------------------------------------------------|
| Radishes with potassium in controlled drought not watered daily | Radishes without potassium not watered daily to induce drought |
| Radishes with potassium watered daily                           | Radishes without potassium watered daily.                      |

## Results

After spending the summer with Doctor Craig Kohn at Waterford's 4H club, I measured and watered the radishes accordingly. At the end of the summer I dried out each crop and measured the dry weight of each specimen, about 100 samples from each sample area. What this proves is that the radish addition to the soil did promote crop growth yield during a stress environment.

## Impact

This is important research because a reported nearly 40% of America's land endures some sort of a drought according to NOAA and that includes areas without growth potential. My research proved that the addition of potassium to soil can increase the crop yields experiencing a stress of water withdrawal and overall increase America crop yields. Research was presented to Runzheimer's International on behalf of a 1,500 student body and scholarship to apply towards higher education. The subject touches a billion dollar industry and led to more than a 125% in academic funding.

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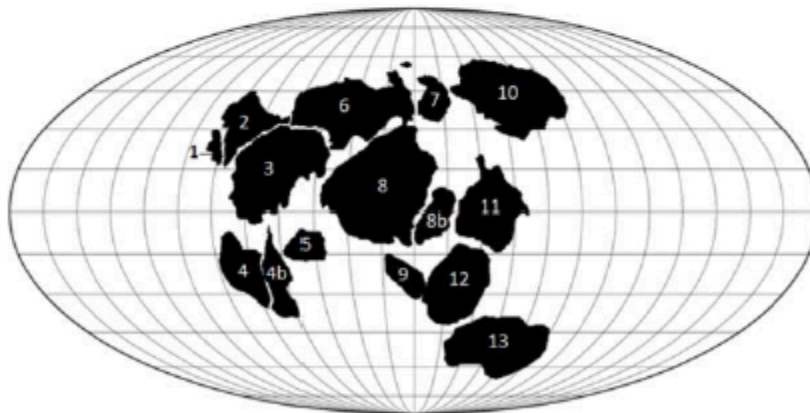
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# COMPUTER MODELING RESEARCH

## 1. Introduction and Background

The Neoproterozoic Era is defined in geologic time from ~1,000 to ~541 million years ago. The Neoproterozoic is often characterized by researchers as a duration of the earth's history in which global systems dramatically shifted and conditions necessary for the emergence of more complex life became prevalent. This period of time underwent many critical events that contributed to defining present day earth. Notable events in this era include the assembly of the supercontinent Rodinia

(Figure 1.) (marking the start of the Tonian period), along with the subsequent disassembly, and the last stages of the Great Oxygenation Event, during which atmospheric oxygen began accumulating in earnest and marked the start of the Cryogenian period (*Chandler and Sohl, 2000*). The most remarkable findings from the Neoproterozoic Era include the geologic and the paleomagnetic evidence that suggest multiple widespread glaciation events that extended to low latitudes, even reaching near equatorial latitudes at some



**Figure 1.** Mollweide map projection of the Rodinia paleocontinental configuration during the Sturtian Glacial Interval (~750Ma). Cratons are as follows: 1, Madagascar; 2, India; 3, East Antarctica; 4, 4b, Congo-Sao Francisco; 5, Kalahari; 6, Australia; 7, South China; 8, 8b, Laurentia; 9, Rio Plato; 10, Siberia; 11, Baltica; 12, Amazonia; 13, West Africa.

points (Harland and Bidgood, 1959; Harland, 1964; Spencer, 1971; Li, et al. 1995; Fairchild, et al., 2018).

Kirschvink first termed the phrase Snowball Earth in his 1992 hypothesis. This hypothesis states that at some point in Earth's history, the globe underwent total glaciation, (Harland, 1964; Kirschvink, 1992). The Snowball Earth Hypothesis was proposed as a means to explain evidence of sea-level glacial deposits found at low paleolatitudes (Harland and Bidgood, 1959; Harland, 1964). The onset of the Sturtian glaciation (750-700 Ma) is thought to have been forced by the initial breakup of Rodinia, marked by when the Laurentia craton first began to separate near the center of the continent (Gernon, et al., 2016). Formerly landlocked regions were then exposed to ocean moisture. This ocean moisture contributed to an enhanced mechanical weathering, which in turn

strengthened chemical weathering. Atmospheric carbon dioxide is removed from the atmosphere via chemical (silicate) weathering, which resulted in a rapid drop in temperature. Declining temperatures helped generate an insuppressible cooling effect which engulfed the entire planet. This process has been termed as a runaway ice albedo feedback loop (Hoffman and Schrag, 2000). Others who have investigated this event have suggested that the placement of the primary land mass, Rodinia, in low latitudes is a vital step in inducing the feedback loop with the potential to bring global glaciation. Ice coverage in high latitudes prevents chemical erosion of rocks beneath the surface and suppresses the carbon burial process. Carbon dioxide in the atmosphere therefore prevents glacial ice sheets from advancing (Schrag, et al., 2002). But since land masses concentrated in the low latitudes rather than high,

chemical erosion was not stifled and the threshold for a runaway freezing episode was soon reached (*Hoffman and Schrag, 2000*). Snow and ice, having high albedos, reflected warm solar insolation from the continents and oceans back into space. Their albedo effect and increase in surface area induced a continuous cycle of global cooling. If ice extent ever reached 50% of the surface area of the planet, it is believed that the feedback would become self-sustaining (*Schrag, et al., 2002*). The Snowball Earth Hypothesis claimed that the earth did in fact reach the 50% threshold necessary to allow for ice to extend all the way to the equator (*Kirschvink, 1992; Hoffman, et al., 1998; Hoffman and Schrag, 2000*).

There are issues with the Snowball Earth Hypothesis, however. One major aspect that was slightly overlooked and needed further exploring was a mechanism for

deglaciation. As stated in *Hoffman, et al. (1998)*, the cooling feedback would become self-sustaining should the majority of the earth's surface undergo glaciation. In order to reverse the work done by the cooling feedback loop, the Earth would require massive amounts of radiative input in order to facilitate its escape from the feedback loop (*Crowley, et al., 2001*). This deglaciation conundrum is evaded, though, if the planet never reached a full Snowball state in the first place where ice reached the equator. A Waterbelt (*Pierrehumbert, et al., 2011*) state has been proposed to counter the idea of equatorial glaciation and elaborates that a band of water existed encompassing the equator. This Waterbelt state is favored by current studies and researchers for two reasons: (1) even with the most up to date Rodinia continental configuration and extremely low levels of atmospheric CO<sub>2</sub>, a solid Snowball state is

extremely difficult to induce in models without setting an unrealistically low solar luminosity (Crowley and Baum, 1993; Chandler and Sohl, 2000; Tziperman, et al., 2012) and (2) there is a lack of evidence that a major extinction event occurred. If total glaciation occurred, then the oceans would be completely sealed and there would be evidence for a widespread elimination of photosynthetic plankton in the stratigraphic record (Tziperman, et al., 2012; Rose, 2015).

There are two primary goals for this paper. The first goal is to determine whether CO<sub>2</sub> levels or solar luminosity have a greater effect on glacial dynamics. Secondly, this paper strives to determine what basic forcings might be necessary to induce total glaciation.

## **2. METHODS**

### **2.1. The Model**

EdGCM (Chandler, et al., 2005) is a general circulation model based on the Goddard Institute for Space Studies general circulation model (Hansen, et al., 1983; Chandler and Sohl, 2000), but is modified for academic purposes. It is a three dimensional model which solves numerically the physical conservation equations for mass, energy, momentum, and moisture, along with equation of state for 8°x10° grid spaces that extend nine layers in the atmosphere. A parsed earth allows for calculations to be made on specific regions and helps the user identify specific areas of interest. Both a seasonal and a 24-hour diurnal solar cycles are factored in to make temperature calculations. Snow accumulation accounts for the combination of snowfall, melting, and sublimation. Ocean heat convergences account for seasonal and regional variances in climate. Surface energy fluxes are model

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derived values; both the ocean heat convergences and the surface energy fluxes are used to calculate sea surface temperatures.

The GCM uses five harmonics when defining a seasonally varied energy flux and upper ocean energy storage to facilitate more accurate approximations in regions of sea ice formation. Sea ice forms via solely thermodynamic mechanisms with no influence from wind stress or ocean dynamics. When a grid cell reaches  $-1.6^{\circ}$ , a block of ice 0.5m thick will form over a fraction of the cell. This ice will respond as needed to maintain energy balance; this includes growing or shrinking both vertically

and horizontally. Ice will begin to melt when a grid cell warms to  $0^{\circ}\text{C}$ . There is a simulated ice/albedo feedback that is relevant to this study as it would alter sea ice and snow cover and is fully explained in *Hansen, et al. (1983)*. Essentially, the albedo of snow is a function of age and depth, with fresh snow maintaining an albedo of 0.85 for 50 days and changes to 0.5 after 50 days. The albedo of snow-free sea ice is set to 0.3 in the near infrared and 0.55 in the visible and is independent of age and thickness.

## 2.2. Forcings

This study aims to determine the dependency of glaciation on luminosity

**Table 1.** Values for Altered Forcings

| Run Label | CO2 Concentration (ppm) | Solar Reduction (%) |
|-----------|-------------------------|---------------------|
| A         | 120                     | 4                   |
| B         | 140                     | 6                   |
| C         | 160                     | 8                   |
| D         | 120                     | 4                   |
| E         | 140                     | 6                   |
| F         | 160                     | 8                   |
| G         | 120                     | 4                   |
| H         | 140                     | 6                   |
| I         | 160                     | 8                   |

Forcings that were altered for each simulation are shown; other forcings remain the same as Run A. All simulations use the Sturtian paleocontinental reconstruction (see Figure 1.) as a land surface boundary condition

versus atmospheric carbon dioxide concentrations. Nine trials ran using the Rodinia configuration (Figure 1.) as the boundary conditions. Rodinia was given a uniform desert condition with no plants present, as vascular vegetation is undeveloped at this time (*Chandler and Sohl, 2000*). The trials ran using a combination of varied CO<sub>2</sub> concentrations and lowered solar luminosity, as listed in Table 1. Orbital forcings were based on modern conditions, with obliquity set to 23.44°, eccentricity set to 0.0167, and omegaT (precession) set to 282.9°. Additionally, Earth's rotation rate was set to the modern rate of 24-hours per 1 rotation, rather than an increased 18-hour rotation period that would have been seen in the Neoproterozoic (*Hyde et al., 2000*).

We hypothesized that despite drastic reductions in both solar luminosity and CO<sub>2</sub> concentrations, EdGCM would not be able

to model a total glaciation without updated ocean dynamics and the feature to model ice sheets. We also hypothesized that solar luminosity would have a greater effect on the extent of sea ice and snow coverage.

### 3. RESULTS

All of the simulation ran long enough for the earth's climate to reach an equilibrium point, which varied from trial to trial, but generally occurred at or around 50 years. After equilibrium was established, the model continued to run each simulation for approximately 10 years, after which the physics of the system failed. Some of the runs (A, B, and C) ran nearly to completion, while other runs failed after only 60 years. This will be further discussed in section 4.1.

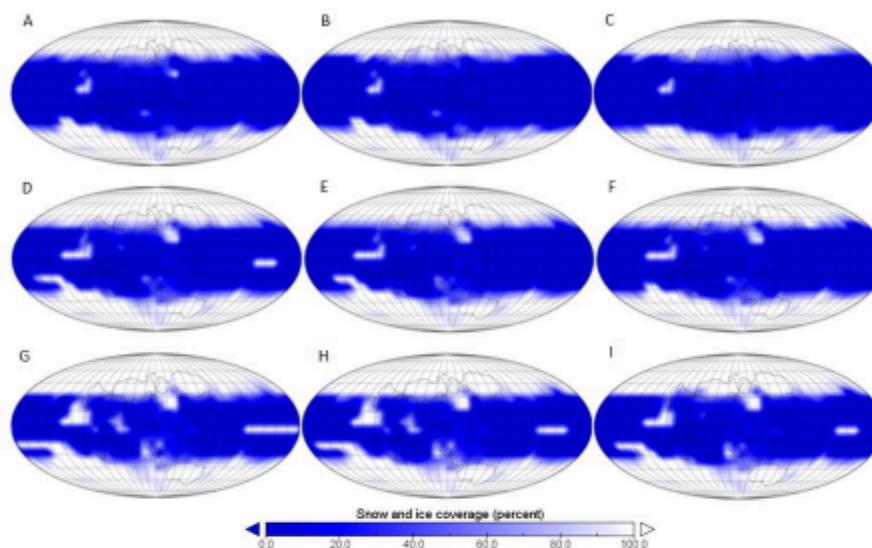


Figure 2. Snow and ice coverage. Individual maps are labeled according to the run label. All simulations use the Sturtian paleocontinental reconstruction (Figure 1.).

### 3.1. Snow and Ice Cover

Run C proved to be the warmest of all nine trials, with an annual average of 38.1% snow and ice cover. Coverage did not extend past 32° latitude. Runs D, E, and F had an average combined snow and ice surface coverage of 45.8%. Of these three runs, snow and ice coverage in run D spread to the lowest latitudes, reaching 31° latitude in both the northern and southern hemispheres. Each hemisphere also showed

ice tongues that approached nearly 25° latitude, but no further. Runs G, H, and I resulted in the earth's coldest conditions with an average combined snow and ice coverage of 52.23%. Runs G and H surpassed the 50% snow and ice coverage marker, while run I was fairly close. The run nearly reached the critical 50% coverage required to launch a self-sustaining feedback loop. Run I displayed an average snow and ice coverage of 49.6%. Run G



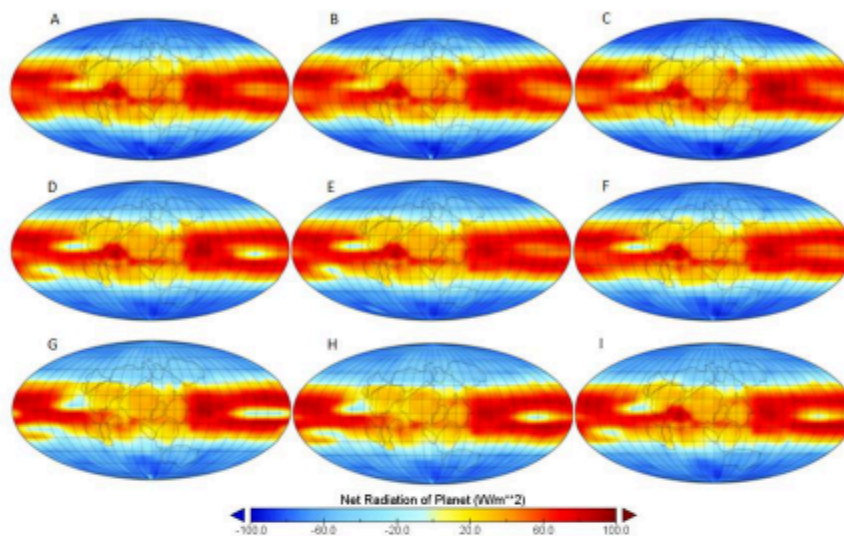


Figure 3. Annual net radiation (see Figure 2 for legend).

proved to be the coldest run with snow and ice coverage reaching to nearly 28° latitude. This run also experienced cold tongues that extended down to 25° latitude.

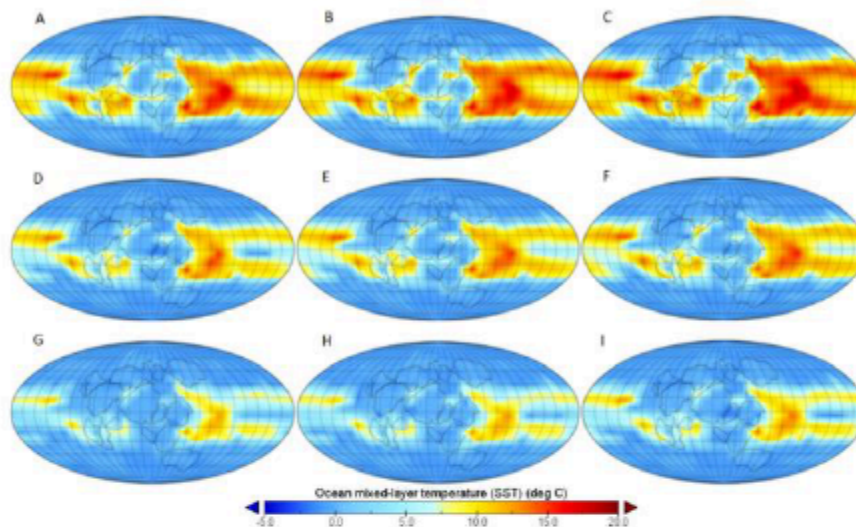
### 3.2. Annual Net Radiation

The net radiation for all nine runs nearly approached zero. Only runs B and C displayed positive net radiations which ended up being valued at 0.3 W/m<sup>2</sup> and 0.5 W/m<sup>2</sup> respectively. The other seven remaining runs experienced an average net

global cooling effect. Run G generated the strongest net radiative cooling at -3.0 W/m<sup>2</sup>.

### 3.3. Annual Sea Surface Temperatures

Sea surface temperatures were reasonable and correlate with the results for snow and ice coverage and net radiation. The warmest temperatures occur near the equator and progressively get colder toward the poles. In the warmest scenario (C), the mean sea surface



**Figure 4.** Annual sea surface temperatures (see Figure 2 for legend).

temperature was 5.1°C. This run had a belt of much warmer water in the low latitudes, with a maximum of 18°C. The equator to pole gradient in the coldest scenario (G) is almost nonexistent, with the majority of the oceans residing at 1.8°C and isolated areas near the equator reaching only 13.2°C.

### 3.4. Cloud Cover

The distribution of clouds varies by both cloud height and latitude (see Figure 4a). Coverage by high clouds in both runs

lingers at around 20% in the mid-latitudes, and rises to around 30% in the tropics. High cloud cover is zero at the poles. Coverage by low clouds reveals more latitudinal variance, reaching up to 80% coverage in the poles, 50-60% in the mid-latitudes, and dropping to about 30% in the tropics.

## 4. Discussion

### 4.1. Errors and Anomalies

As reported above, each run reached equilibrium, but none were able to run the

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simulation to completion. Since EdGCM works mainly using calculations of moisture, mass, energy, and momentum, if one or more of these calculations is unable to compute to completion, the physics of the system fails and the run stops. The simulation is unable to create ice sheets, but is capable of modelling sea ice formation as described in our methods. However, since the simulation uses a mixed-layer ocean model, it is possible that extreme conditions, such as those seen in our simulations, could cause one or more grid cells to freeze solid through the bottom of the layer. The result of grids freezing repeatedly this may ruin the simulation. When this occurs repeatedly, the result is a disruption of the calculations needed to continue running the simulation, and thus the run is terminated at this point (M. Chandler, personal communication, October 10, 2016; M.Chandler, personal

communication, November 15, 2018). Because none of the runs ended at the same time, data from years 55-60 were used to create plots from all runs.

Review of the snow and ice cover revealed more doubt in the EdGCM results. Simulations revealed isolated regions of ice formation near the equator off the coasts of the East Antarctic craton and Baltica. These equatorial glaciations are present in each run, despite forcings that would strongly hinder ice growth at these latitudes. Furthermore, the extent of this anomalous glaciated region does not change from run to run (within runs of the same solar forcings). This piece of evidence was therefore written off as error in the simulation's calculations. This error is also present in net radiation, sea surface temperatures, and cloud cover plots, and is also disregarded. After dismissing this data as credible evidence, the simulations show

suggestion that orbital periods may have been ~20% shorter (Crowley and Baum, 1993). We cannot accurately incorporate these suggestions into models because we do not know the difference in magnitude of the forcings during the time period of interest from the current magnitude.

Orbital forcings were unchanged from current states, thus the distribution of radiation was similar to current distribution of radiation, with a deficit over the poles and surplus over the equator. Comparison of net planetary radiation from both runs C and G, as well as present-day net planetary radiation measurements suggests that latitudinal variability in net radiation increases as luminosity increases (Figure 5A).

#### 4.5. Response of Cloud Cover to Forcings

Cloud cover may have been both an influence and result of net planetary radiation. High clouds generally have a warming effect and a lower water content, while low clouds tend to have a cooling effect and higher water content (Pierrehumbert, et al., 2005; Abbot, et al., 2012). The cooling and warming effects align with the distribution of net radiation seen in the simulations (Figure 5). The

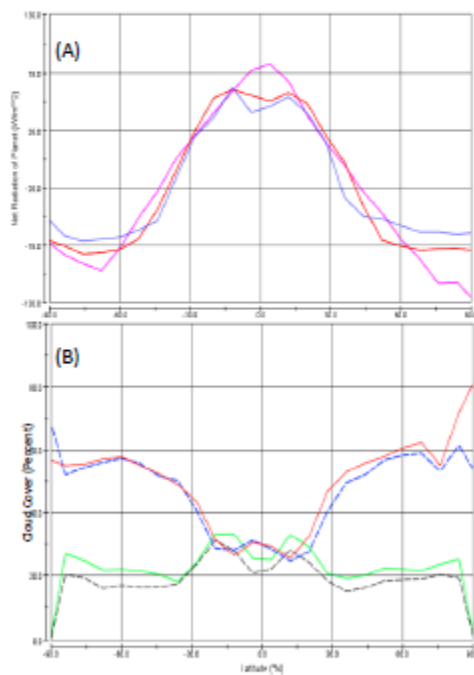


Figure 5. (A) Annual net planetary radiation from runs C (blue), G (red), and modern measurements (pink). (B) High and low cloud coverage from run C (green and blue dashed, respectively) and from run G (black dashed and red, respectively).

tropics receive the most coverage from high clouds seen anywhere in the simulation, while also seeing the scarcest coverage from low clouds. Clouds in the tropics are thus believed to have a slight warming effect that affects net radiation and extent of snow and ice cover.

#### 4.6. Comparison to the Literature

Our results closely resemble those of previous modelling experiments performed to look at different regions and times of glaciation periods on earth (Crowley and Baum, 1993; Jenkins and Frakes, 1998; Chandler and Sohl, 2000).

Crowley and Baum (1993) initially used a simple southern hemisphere continent consistent with the basic Pannotia/Varanger configuration (Figure 6, taken from Chandler and Sohl, 2000) to model the Varanger glaciation (~610-575 Ma), and secondly used a simple rectangular continent in the low latitudes to

simulate the Sturtian glaciation. Their second model's results showed that glaciation could initiate on a horizontally oriented low latitude continent with CO<sub>2</sub> levels at 120 ppm, but requires a 20% reduction in solar luminosity. The 20% reductions are unrealistic conditions for earth to sustain when interpreting the solar luminosity curve (Figure 7, from Crowley and Baum, 1993). The simulation also showed that a positive snow and ice albedo feedback loop can be induced over a region with permanent snow, which can influence and encourage mid-to-high latitude glaciation. Crowley and Baum (1993) satisfied many pieces of evidence for low latitude glaciation, but were unable to

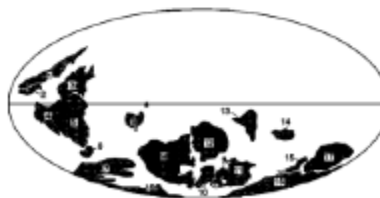


Figure 6. Pannotia/Varanger configuration (Chandler and Sohl, 2000)

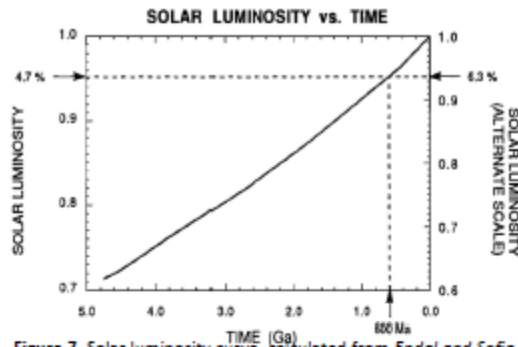


Figure 7. Solar luminosity curve, calculated from Endal and Sofia, 1981.

reveal a mechanism for sea-level equatorial glaciation (Crowley and Baum, 1993).

Jenkins and Frakes (1998) used the GENESIS R15 model (developed by Johnson and Pollard, 1995) with a horizontal resolution of  $4.5^{\circ} \times 7.5^{\circ}$ , to run a series of simulations with forcings similar the ones used in our study. Additional boundary parameters were included in this case. GENESIS ran with a simple horizontally oriented rectangular continent, centered on the equator that included a mountain range running north-to-south across the face of the continent. The results of the simulations showed that it is difficult to produce

conditions favorable for low-latitude glaciation. Similar to Crowley and Baum (1993), the GENESIS simulations showed that initiation of global glaciation occurs over land where permanent snow occurs. Jenkins and Frakes also observe, however, that

arid conditions combined with cloud cover at less than 20 percent over the interior of the continent inhibits the possibility of equatorial glaciation. Jenkins and Frakes thus argue that incident solar radiation at the surface must decrease and precipitation rates must increase to accommodate for the conditions necessary to form a total global glaciation (Jenkins and Frakes, 1998).

Chandler and Sohl (2000) use the GISS GCM model that is the base model of EdGCM. Their study, however, focused on the Varanger glaciation rather than the Sturtian glaciation and the only parameter changed from the ones used in this study (in

the runs that are analogous to ours) was the continental configuration (Figure 6). Though the continental boundaries were shifted, the GCM was unable to produce a 'hardball' state where the entire globe was covered in ice. The results of their simulations produced cloud cover that was similar in value to that observed in this study, though the distribution was shifted due to their boundary conditions (*Chandler and Sohl, 2000*).

### 5. Conclusion

The EdGCM model was designed as a way for students to explore climate variables and dynamics. For this purpose, the model is enough in its calculations and calibrations that would be relevant for academic purposes. With EdGCM, we were able to simulate various scenarios with mixed forcings to suggest that the climate system is more sensitive to changes in solar luminosity than to changes in atmospheric

CO<sub>2</sub> concentrations. From this conclusion, we also suggest that further trials should be run at 6% reduction in solar luminosity with decreasing amounts of CO<sub>2</sub> as an attempt to induce total glaciation.

However, there are certain parameters that are vital to the investigation of climate states, past and present, that are absent from the model, but must be addressed. These include, but are not limited to, ice sheet dynamics, cloud mechanics and feedbacks, ice-albedo feedbacks, orbital and diurnal variations, and seasonal changes in convection cells. Efforts have been made over the past few years to provide more accurate models, but given the complexity of the climate system, progress has been slow. Major steps have been taken, however, in experimenting with different glaciation and deglaciation mechanisms. *Abbot et al. (2010)*, *Abbot, et al. (2012)*, and *Abbot (2014)* are a series of

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papers that investigate and make significant headway on quantifying cloud radiative forcing and deglaciation thresholds. CO<sub>2</sub> forcings are difficult to model, as one must account for the neutralization of the carbonate reservoir (Crowley *et al.*, 2001, Pierrehumbert, 2011), which, of course, leads to the need to account for the approximate size of the carbonate reservoir. Though, Abbot (2010, 2013, and 2014) has made significant headway in this topic, there are more factors that must be investigated.

Progress has been made in the reconstruction of the paleogeography. Arnaud (2004) performed sedimentological analysis of the Port Askaig Formation, a formation that is part of a Supergroup that is thought to record the rifting of Rodinia as well as containing 'cap' carbonates that are characteristic of the rapid Sturtian deglaciation. The critical piece of evidence

from this study is the assessment of cross-bedding in the sandstone member of this formation. Analysis highly suggests that this cross bedding formed via tidal currents, i.e. the ocean was not sealed during the Sturtian glaciation (Arnaud, 2004).

Furthermore, strides have been made in the field of paleomagnetism, which help to further parameterize the boundary conditions of the Rodinia continental configuration (Tziperman, *et al.*, 2012; Fairchild, *et al.*, 2018; Zhao, *et al.*, 2018; Campanha, *et al.*, 2019).

And lastly, there have been contributions to the field of paleoceanography, which further add to the list of parameterizations that our models are built upon. Ashkenazy, *et al.* (2013), among several others, has taken initial steps in simulating the dynamics of a Snowball Earth ocean. Their initial model failed to include chemically driven density differences. Gernon, *et al.*



(2015) uses Monte Carlo simulations to demonstrate how enhanced volcanism due to the Rodinia break up contributes to an alkaline shift in ocean chemistry before and during the Sturtian glaciation. This, combined with paleogeographic data, will greatly help further our understanding of the ocean circulatory patterns. Developing a more complete understanding of the ocean process will be relevant in minding their impact on glaciation.

As a scientific community, we are moving toward a more complete understanding of climate dynamics and paleoclimate. Our knowledge of the climate dynamics and paleoclimate will allow for us to answer even bigger questions and better prepare for any future advancements. Future studies will hopefully be able to accurately incorporate this burgeoning information into new, more precise models. More accurate information will then be more

reliable and more mainstreamed in everyday applications.

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# AIR QUALITY INVESTIGATION

```
# http://www.acgeospatial.co.uk/sentinel-5p-and-python/

import netCDF4

from netCDF4 import Dataset

import numpy as np

import glob

import pandas as pd

import matplotlib.pyplot as plt

from mpl_toolkits.basemap import Basemap

import os
```

[1] Python

```
path = 'C:\\Users\\akiec\\Downloads\\October\\'
for filename in glob.glob(os.path.join(path, '*.nc')):
    fh=netCDF4.Dataset(filename, 'r')
    print(fh)

##filelist= glob.glob(r'C:\\Users\\akiec\\Downloads\\Weekend*.nc')
##fh = Dataset(filelist,mode = 'r')
#my_nc_file= (r"C:\Users\akiec\Downloads\Weekend\S5P_OFFL_L2__TroposphericColumnTotalOzone.nc")
#fh = Dataset(my_nc_file,mode = 'r')#create variable that reads
#variable will be a list of files
#create a new variable with y dimension average
```

[2] Python

```

... <class 'netCDF4._netCDF4.Dataset'>
root group (NETCDF4 data model, file format HDF5):
  Conventions: CF-1.7
  institution: KNMI
  source: Sentinel 5 precursor, TROPOMI, space-borne remote sensing, L2
  history: 2018-11-01 18:56:41 f_s5pops tropnll2dp /mnt/data1/storage_offl_l2/cache_offl_l2/WORKING-564737083/JobOrder.564737059.xml
  summary: TROPOMI/S5P Aerosol Index 1-Orbit L2 Swath 7x3.5km
  tracking_id: 9a56bd48-541b-4c2d-871d-abc21b9498f7
  id: S5P_OFFL_L2_AER_AI_20181026T193249_20181026T211419_05369_01_010200_20181101T185257
  time_reference: 2018-10-26T00:00:00Z
  time_reference_days_since_1950: 25135
  time_reference_julian_day: 2458417.5
  time_reference_seconds_since_1970: 1540512000
  time_coverage_start: 2018-10-26T19:54:24Z
  time_coverage_end: 2018-10-26T20:52:47Z
  time_coverage_duration: PT3503.000S
  time_coverage_resolution: PT1.080S
  orbit: 5369
  references: http://www.tropomi.eu/data-products/aerosol-index
  processor_version: 1.2.0
  keywords_vocabulary: AGU index terms, http://publications.agu.org/author-resource-center/index-terms/
  keywords: 0300 Atmospheric Composition and Structure; 0305 Aerosols and Particles; 0360 Radiation, Transmission and Scattering; 3311 Clouds and Aerosols;
  standard_name_vocabulary: NetCDF Climate and Forecast Metadata Conventions Standard Name Table (v29, 08 July 2015), http://cfconventions.org/standard-name
  naming_authority: nl.knmi
  cdm_data_type: Swath
...
  dimensions(sizes):
  variables(dimensions):
  groups: PRODUCT, METADATA

```

Output is truncated. View as a [scrollable element](#) or open in a [text editor](#). Adjust cell output settings...

```

lon = fh.groups['PRODUCT'].variables['longitude'][:,0,::]
lat = fh.groups['PRODUCT'].variables['latitude'][:,0,::]

no2 = fh.groups['PRODUCT'].variables['nitrogen_dioxide_tropospheric_column_precision'][:,0,::]

print(np.mean(no2))

```

[3] Python

```

... 7.978506436561867e-06

```

```

no2_units = fh.groups['PRODUCT'].variables['nitrogen_dioxide_tropospheric_column_precision'].units

```

[4] Python

```

from matplotlib.colors import LogNorm

```

[5] Python

```

lon_0 = lon.mean()
lat_0 = lat.mean()

```

[6] Python

```

m = Basemap(projection='cyl', resolution='l',
            llcrnrlon=-115, llcrnrlat=30, urcrnrlon=-108, urcrnrlat=38) #this is where you can zoom as desired

```

[7] Python

```
m = Basemap(projection='cyl', resolution='l',
            llcrnrlon=-115,llcrnrlat=30,urcrnrlon=-108,urcrnrlat=38) #this is where you can zoom as desired

xi, yi = m(lons, lats)

# Plot Data

cs = m.pcolor(xi,yi,np.squeeze(no2),norm=LogNorm(), cmap='jet')
# Add Grid Lines

m.drawparallels(np.arange(-80., 81., 10.), labels=[1,0,0,0], fontsize=10)

m.drawmeridians(np.arange(-180., 181., 10.), labels=[0,0,0,1], fontsize=10)

# Add Coastlines, States, and Country Boundaries

m.drawcoastlines()

m.drawstates()

m.drawcountries()
#map (lat, long) to (x,y)
x,y = m(-112.0740, 33.4484)
plt.plot(x,y, 'ok', markersize=2)
plt.text(x,y, 'PHX', fontsize=12)

cbar = m.colorbar(cs, location='bottom', pad="10%")

cbar.set_label(no2_units)

# Add Title
```

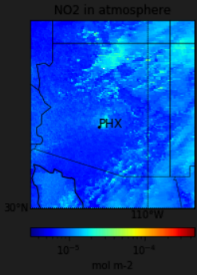
[7] Python

```
# Add Title

plt.title('NO2 in atmosphere')
plt.show()
```

[7] Python

...



NO2 in atmosphere

30°N 110°W

10<sup>-5</sup> 10<sup>-4</sup> mol m<sup>-2</sup>



## **Summary**

This research project aims to use satellite measurements to study nitrogen dioxide ( $\text{NO}_2$ ) in the troposphere over Arizona. Furthermore, the study initially aimed to study the relationship between traffic relationships and  $\text{NO}_2$  concentrations over Phoenix, AZ. As the study progressed, it evolved to focus on a relationship between wind and  $\text{NO}_2$ .

## **Motivation for Study**

Nitrous oxides ( $\text{NO}_x$ ), composed of  $\text{NO}_2$  and nitrogen oxide ( $\text{NO}$ ), is an ingredient in the formation of tropospheric ozone ( $\text{O}_3$ ). Nitrous oxides are highly reactive gases that, when in the presence of sunlight, can form  $\text{O}_3$ . (US EPA, 2016).

Ozone in the stratosphere blocks ultraviolet rays from penetrating deep into the earth's atmosphere. This is  $\text{O}_3$  that protects human health. When  $\text{O}_3$  forms in the troposphere, it becomes a pollutant that provokes human health issues. In the troposphere, people can directly inhale  $\text{O}_3$  which may irritate airways and damage the human respiratory system (US EPA, 2016).

Aside from its contribution to  $\text{O}_3$  formation,  $\text{NO}_2$  alone is threatening to human health. The United States Environmental Protection Agency (EPA) explains that preexisting respiratory diseases, such as asthma, can become aggravated in the presence of  $\text{NO}_2$ . Severe cases of exposure can even lead to hospitalization (US EPA, 2016). Because  $\text{NO}_2$  is so harmful to human health and promotes tropospheric  $\text{O}_3$ , we need to increase our understanding of our contributions to  $\text{NO}_2$  emissions and how we can mitigate its damaging impacts.

If we refine our understanding of NO<sub>2</sub> sources, then we can extend our knowledge of tropospheric O<sub>3</sub>. As an increasingly important dialogue about climate change continues to circulate around the public, we need to continue to improve our own understanding of air pollution, meaning we need more reliable and exact data to tell us a more complete and accurate story about what is happening between anthropogenic activity and the atmosphere.

### **Abstract**

As one of the six criteria pollutants, defined by the EPA as air pollutants that harm human health, NO<sub>2</sub> has been studied to understand its origins and lasting impacts on the atmosphere (US EPA, 2016). A major source of tropospheric NO<sub>2</sub> comes from fuel burning sources such as vehicle emissions, power plants, and off-road equipment (US EPA, 2016). A study from Lund University found a relationship between rush-hour traffic and NO<sub>2</sub> levels in the breathable air. (Science Daily, 2018) On road vehicle traffic peaks during rush hours, therefore NO<sub>2</sub> emissions peak in rush hour traffic.

In July 2004, NASA launched the Ozone Monitoring Instrument (OMI) into space to orbit the earth. The device was designed to study atmospheric chemistry and dynamics. OMI has the capability to distinguish between different aerosols in the atmosphere and reveal distinctions in atmospheric composition (Earth Online, 2019). Following the launch of OMI was the launch of the Tropospheric Monitoring Instrument (TROPOMI) in June 2018. Ever since the satellite's commission, it has been orbiting the earth to collect measurements of atmospheric species,

climate observations, and O<sub>3</sub> layer activity (TROPOMI, 2018). TROPOMI provides a more complete data set of atmospheric measurements and has a higher resolution display of atmospheric composition than OMI.

Wind impacts atmospheric chemistry by displacing atmospheric species from their sources.

Weak winds result in stagnant air. When NO<sub>2</sub> sources pollute stagnant air spaces, the pollution can become highly concentrated over a small area. Whereas stronger winds disperse air pollution from its sources and can impact a larger area in lower concentrations. Wind direction dictates where the air pollution goes (National Weather Service, 2017).

### **Introduction**

Wind can reduce or increase air pollution activity over large areas. Winds that blow over a source of air pollution can disperse the winds over a greater area, allowing for the pollution to decrease over the source area while increasing the amount of the pollutant over areas that would otherwise not have the pollution in its airspace. Measuring both winds speed and direction are essential in tracking its relationship to a pollutant in the air (Waikato Region Council).

Urban areas have a large amount of on road vehicles especially in rush-hour traffic. Nitrogen dioxide reacts with volatile organic compounds (VOCs) in the presence of sunlight to form tropospheric ozone. Sources of VOCs include cleaning products, paint, pesticides, and vehicle products (EPA, 2016). These sources of VOCs are typically in the same airspace as the tropospheric NO<sub>2</sub> pollution, where they can react to form O<sub>3</sub>. Not only can NO<sub>2</sub> enable the

development of tropospheric O<sub>3</sub> but also induce asthma and encourage respiratory infections when inhaled (EPA, 2016).

Measurements by TROPOMI provides descriptions about air pollution. With a resolution of 24.5 km<sup>2</sup>, TROPOMI can distinguish specific air pollution details over a single city. In general, the satellite shows a consistency of high NO<sub>2</sub> concentrations to urban areas like Phoenix, with a land area of 1,340 km<sup>2</sup> (World Population Review, 2019, TROPOMI, 2018).

VentuSky, an online application developed by a Czech meteorological company who aims to spread wind and other atmospheric data to the public, uses NOAA archived wind data to show historical wind directions and speeds. These measurements resonate with the relationship of weak winds and lower concentrations of NO<sub>2</sub> sprawled over a larger land area. Wind directions would influence where the NO<sub>2</sub> covers and the wind speeds would gauge the potential distance the pollution could move from its source.

After TROPOMI was launched in October 2017, it performed six months of calibration. Once the calibrations were completed, the satellite commissioned for live usage. Ever since June 2018, the satellite has been collecting atmospheric chemistry and dynamics data (TROPOMI, 2018). The satellite provides detailed coverage over Arizona every single day. A commissioned TROPOMI is great for air pollution studies because it can help write the story of identifying sources of criteria air pollutants and measuring their impacts on human health.

## **Methods**

The first step in this project was to be able to overlay TROPOMI data onto a map. Using Python to write the script, I was able to successfully produce TROPOMI measured data over Arizona. At first, my main interest involved vehicle commuter traffic and its relationship to  $\text{NO}_2$  emissions in Phoenix, Arizona. I wanted to study Phoenix because they have a heavy traffic problem that is only increasing and Phoenix, being in the Valley of the Sun, receives an extreme amount of sunlight each year (ADEQ, 2017). Nitrogen dioxide emissions and sunlight is a recipe for tropospheric  $\text{O}_3$ . Knowing that  $\text{O}_3$  forms in the presence of  $\text{NO}_2$ , it is important to understand the precursors for  $\text{O}_3$  formation to thoroughly understand  $\text{O}_3$  formation.

My initial objective was to reveal a relationship between traffic volume fluctuations and TROPOMI measured  $\text{NO}_2$ . To do this, I needed to plot averaged TROPOMI  $\text{NO}_2$  measurements over 5-day weeks and 2-day weekends. The challenge with averaging TROPOMI data is that right now, NASA's data that is offered is not gridded. Our group in the Center for Sustainability and the Global Environment did not have experience in regridding data. To accomplish this would have added on many more unanticipated weeks of data processing. The data needed to be gridded before being averaged to have an accurate display of averaged  $\text{NO}_2$  measurements. Due to limitations on resources such as time, this project needed to be amended at this point to answer a question that could be answered using single day measurements and forgo the averaged data, at least for now.

After this revision, my next task was to decide what kind of conditions would be optimal in showing weekday and weekend variations in  $\text{NO}_2$  due to commuter traffic. Initially, I used Air Quality Index (AQI) reports from the EPA to find the most severe days of measured  $\text{NO}_2$  over

Phoenix. I thought this approach made sense so that my project could showcase TROPOMI data collection over Phoenix's days of greatest NO<sub>2</sub> pollution concerns. I realized late into my project that this was the opposite approach I should have been using all along. Instead, my research question would be better answered with an analysis of good air quality days over Phoenix. My thought process lead me to believe that if I wanted to see NO<sub>2</sub> potentially originating from vehicle emissions, then a day with poor air quality would have extra factors that would display the NO<sub>2</sub> over all of Arizona, not showing so much of vehicle emission sources, factors like wind moving pollutants across the entire state.

Because of time constraints and the inability to average the TROPOMI measured NO<sub>2</sub> data, this is what my project evolved to. Instead of explicitly addressing vehicle emissions' contributions to NO<sub>2</sub> over Phoenix, I looked at how wind patterns over all of Arizona related to TROPOMI measured NO<sub>2</sub>. VentuSky displays historical and real time weather conditions over the entire world with data provided by NOAA. I was able to find historical data of wind patterns to show a relationship between wind and TROPOMI measured NO<sub>2</sub>.

Another component of this study was to decide not only to study days with low air pollution but a time of the year that made sense. TROPOMI started collecting data in June 2018. From then until now, I decided that a major holiday would cause an abnormal excess of vehicle emissions. Therefore, I started my data analysis with Thanksgiving Day, 22 November 2018.

## **Results**

After evaluating the TROPOMI data with the NOAA wind records, the plots show a consistent relationship with wind speed and direction variances with differences in measured NO<sub>2</sub>. Figure 1 shows TROPOMI measured NO<sub>2</sub> data on the top row and then wind records of their respective days on the bottom row. Phoenix consistently has higher levels of measured NO<sub>2</sub> than the rest of the state of Arizona and consistently lower wind speeds.

The bottom column of figure 1 shows consistently low wind speeds over Phoenix compared to the rest of the state. Areas with lower wind speeds have higher values of NO<sub>2</sub> for that day.

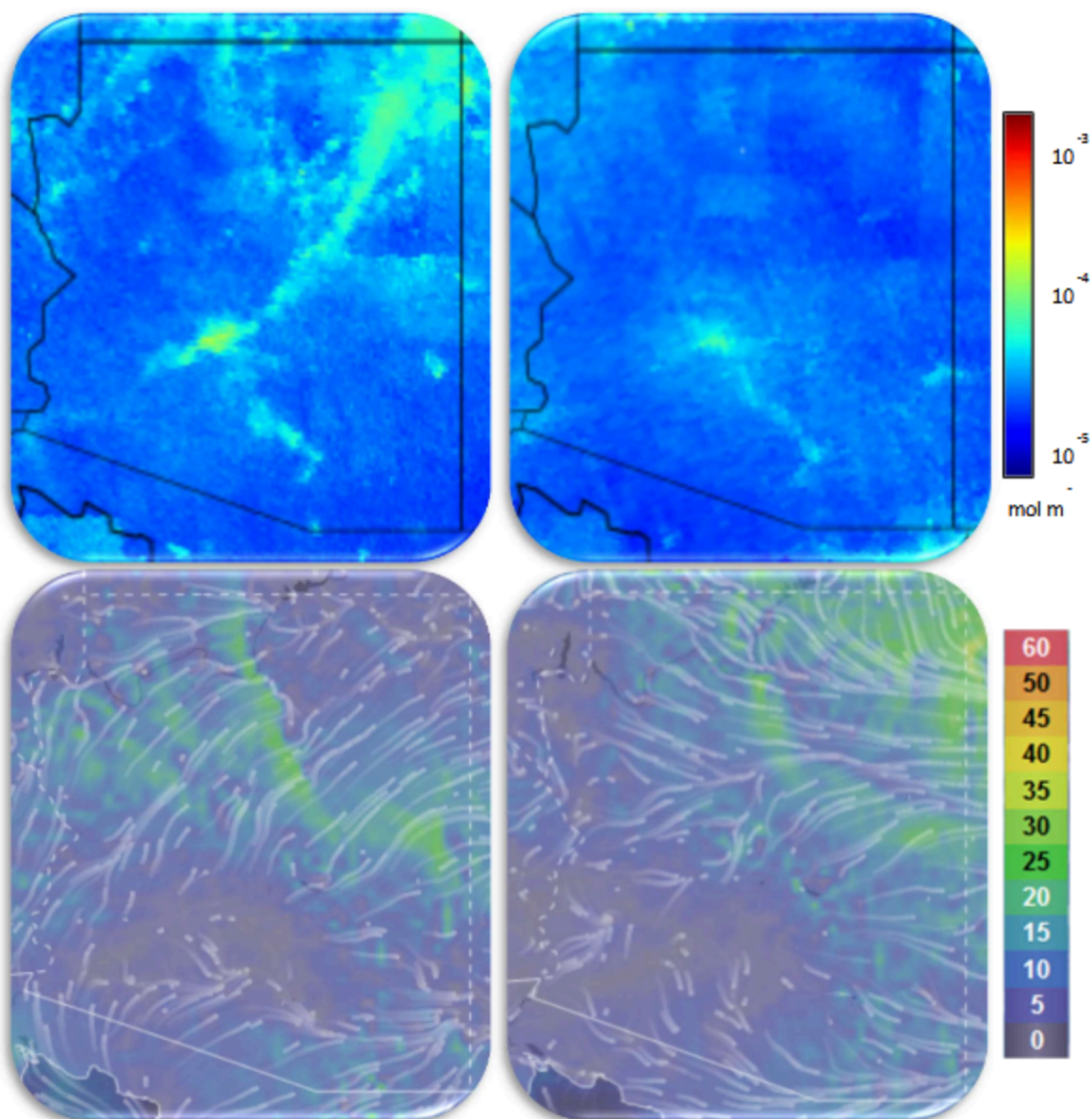
Where and when wind speeds become stronger relates to the increased TROPOMI measured NO<sub>2</sub> concentrations. Based on the NOAA provided wind data and NASA's TROPOMI data archive, figure 1 demonstrates a relationship between wind trends and air pollution variations with a consistent trend of weak winds and high NO<sub>2</sub> over Phoenix.

Although not the final focus of my research, the two days of plots in figure 1 allude to potential vehicle emission impacts on the troposphere. A region of consistently high NO<sub>2</sub> in Arizona is over the Phoenix area. Further, this high NO<sub>2</sub> area is consistent with where there are more densely populated areas and commuter traffic. Figure 1 may reveal the weekday and weekend NO<sub>2</sub> emission variance because the TROPOMI data collected on a Saturday is less severe than the weekday recorded NO<sub>2</sub>, on a Tuesday.

Thanksgiving Day, 22 November has the worst air quality occurrence of all the days studied between 22 November and 29 November 2018. Figure 2 shows the highest occurrence of NO<sub>2</sub> in Arizona over the duration of my study.

Tuesday 27 November

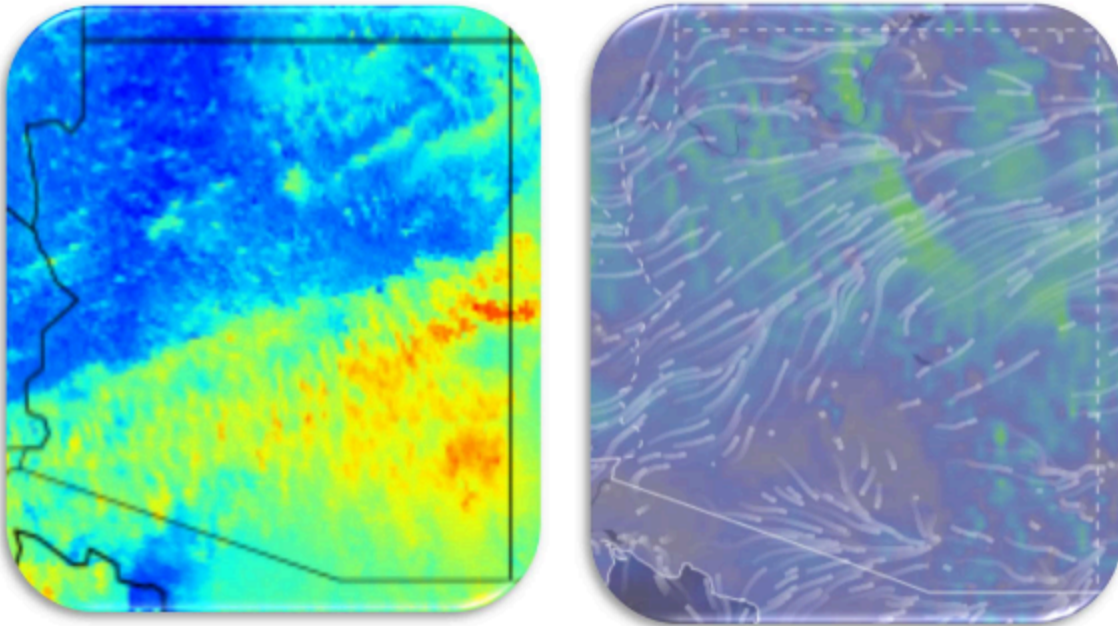
Saturday 24 November



**Figure 1:** TROPOMI measured NO<sub>2</sub> (left) and NOAA recorded winds (right) are compared side by side to show a consistency of relatively high pockets of NO<sub>2</sub> and low wind speeds.



Thursday 22 November



**Figure 2:** The most NO<sub>2</sub> recorded by TROPOMI in this study (left) and same day wind patterns provided by NOAA (right)

### Discussion

Initiated from this project, there are several next steps to take to expand on air quality research to improve public health for not only Arizona residents but also the entire world. One next step I would take would be to isolate my study to Phoenix.

A study of Phoenix would be able to move my project back to its focus, that being able to study a weekday and weekend variance in the amount of NO<sub>2</sub> emissions measured by TROPOMI. In

order to accomplish this, I would change the scale of my project and dedicate the range of NO<sub>2</sub> emissions to only the Phoenix area, including surrounding suburbs.

A key factor to consider with Phoenix is that the city, being in the Valley of the Sun, lies lower than its surrounding area. Because the city is in a valley region, there can be a relationship established to demonstrate how winds are impacted by the Phoenix elevation.

Another piece of the study involves analyzing traffic patterns in the Phoenix area. Traffic trends can reveal places of heavy congestion versus roadways that move with ease. At first a daily log of average traffic trends may be useful, then compare the data to averaged TROPOMI measurements. Averages provide a stronger representation of the bigger picture in Phoenix. Averages remove outlier data and can make a more concrete case of the weekend effect.

### **Conclusion**

Based on the conclusions from figure 1, wind and air pollution have a relationship where the wind influences the concentrations and reach of NO<sub>2</sub>. Every air quality study has unique attributes which need to be considered when investigating the causes and impacts of poor air quality. This study can be expanded to explore additional factors that could impact the presence and behavior of NO<sub>2</sub> over the Phoenix area. Topography, outlying potential sources of additional pollution, and urban planning all could be explored to expand on the knowledge we have of the sources and activity of NO<sub>2</sub> in the area.

Furthermore, not only should NO<sub>2</sub> be studied, but also other criteria pollutants and airborne dust. Phoenix endures dust storm episodes and faces air quality challenges from their own

surrounding land.  $\text{NO}_2$  also reacts with other chemicals in the atmosphere and may have more of an impact beyond its damaging effects on the human respiratory system, such as contributing to the formation of tropospheric  $\text{O}_3$ .

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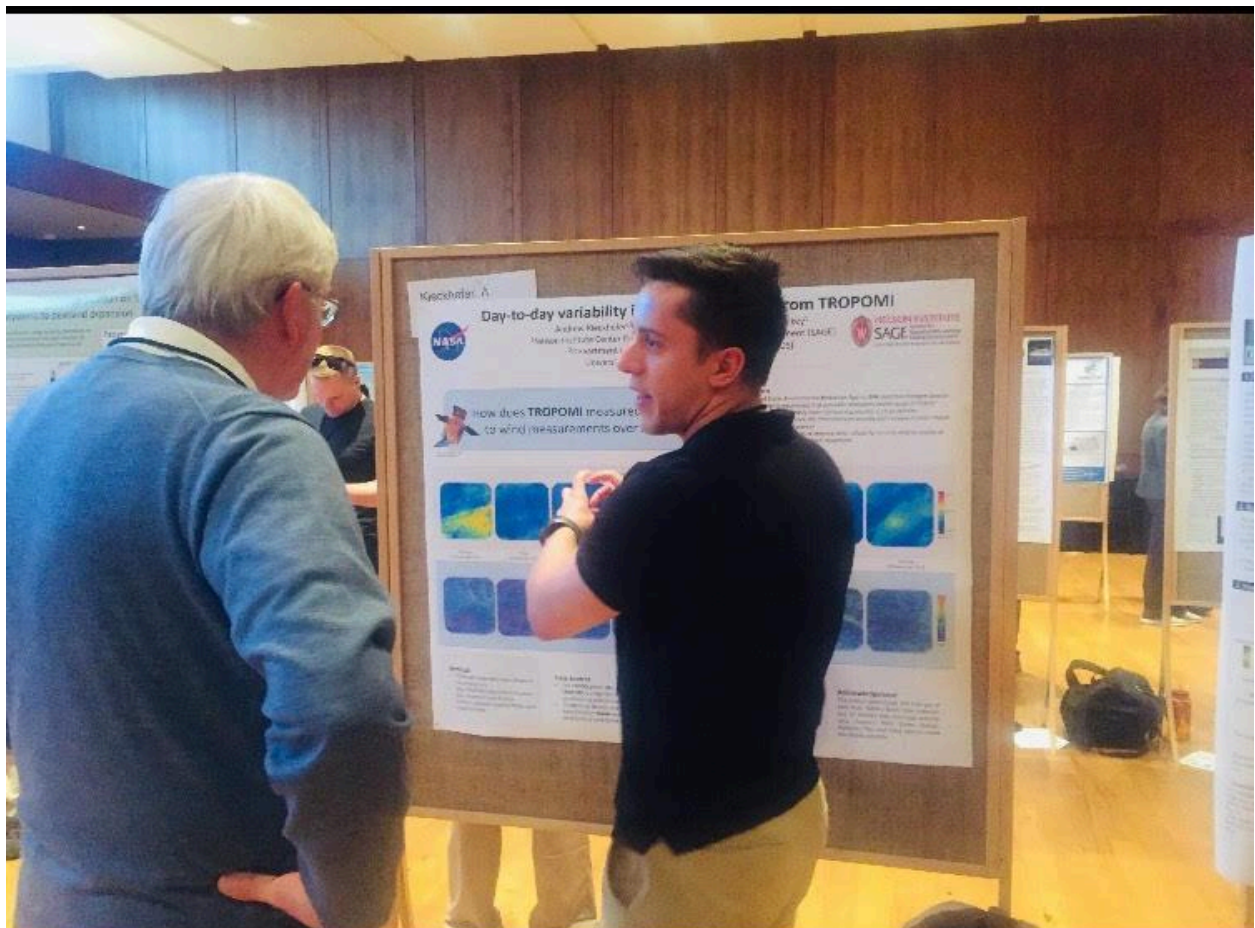
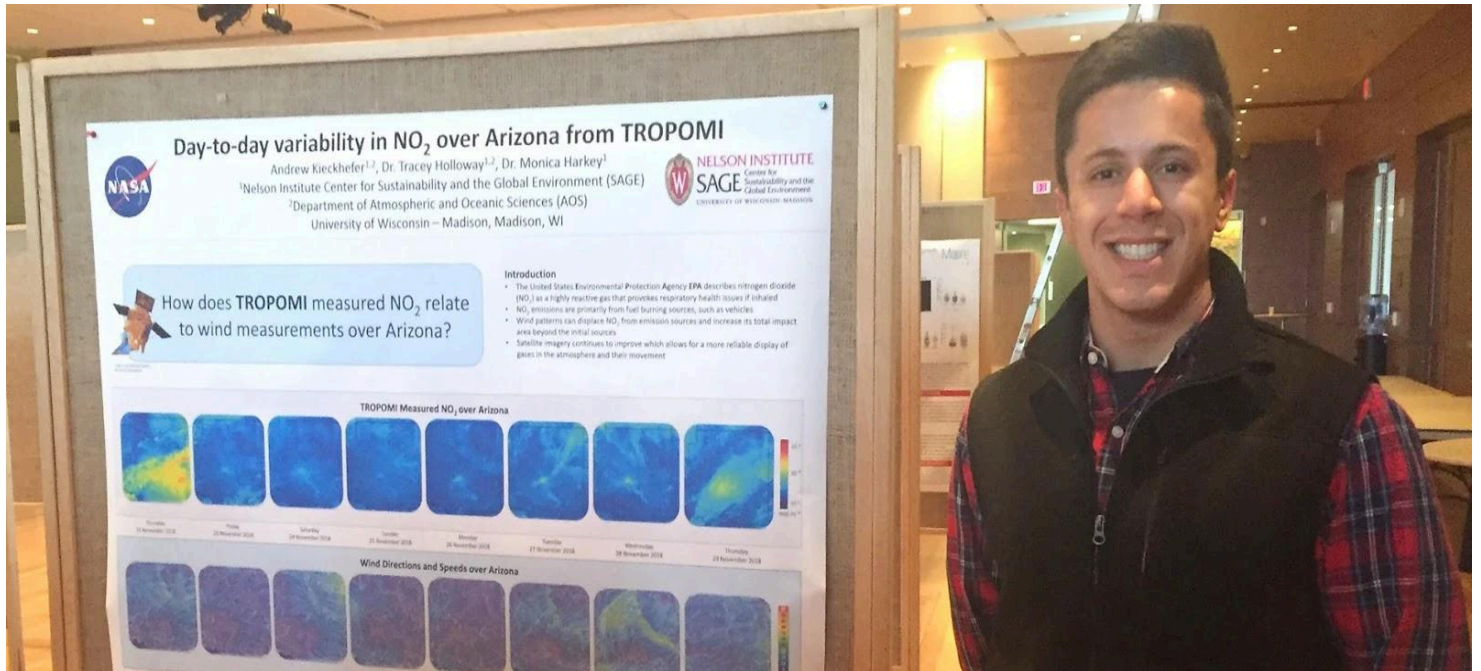
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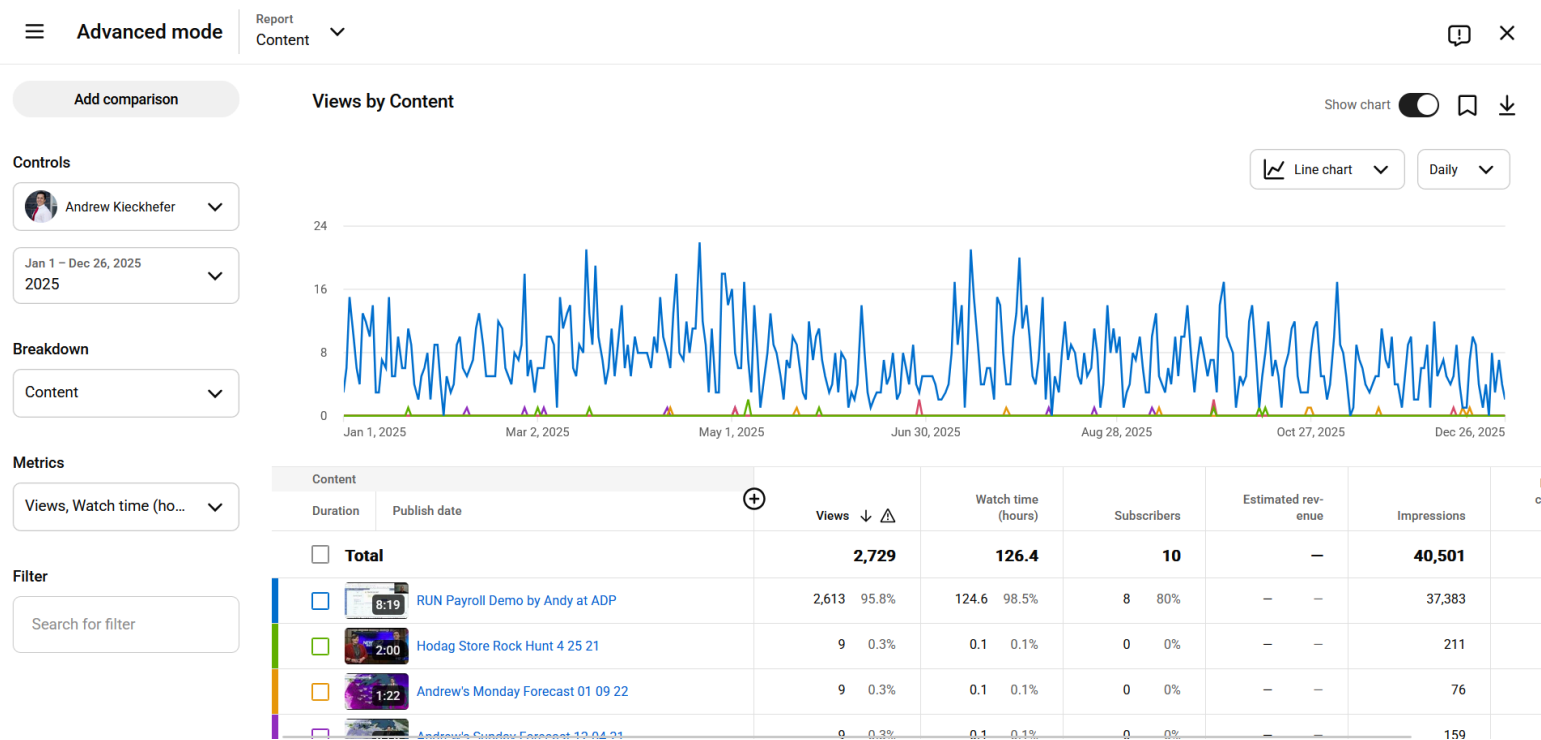
# RESEARCH SYMPOSIUM



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# CREATOR CONTENT



Ongoing research with YouTube Analytics to measure the watching statistics of my news coverage videos compared to my software sales demonstration with 1000% the viewership.. Ability to demonstrate research methods in a fast paced work environment to get the facts out fast and reliably. In fact one feature story I covered was about researching and deploying the story about a local homicide.





### Antigo Homicide Investigation Expanded



Andrew Kieckhefer  
19 subscribers

Analytics

Edit video



0



Share



Ask



Save



Local research on an ongoing murder investigation being reported with the most up to date information. Other features include a building explosion, Wisconsin Department of Natural Resources topics, and other research on local events and happenings. Oh, and of course researching the weather constantly.

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## **AI Mental Health Treatment Services Could Grow the Volume of Patients per Provider. \$2M Funded Study.**

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The hypothesis is that OpenAI could be used in psychotherapy treatment services for mental health. Why does it matter? Unlocking insurance-backed user services could add more stability to the company's financial landscape while increasing the healthcare provider's productivity with more insurance covered patients billed per provider. According to the National Institute for Mental Health, "psychotherapy (also called talk therapy) refers to a variety of treatments that aim to help a person identify and change troubling emotions, thoughts, and behaviors." (Psychotherapies - National Institute of Mental Health (NIMH), 2024). This research is to propose that utilizing artificial intelligence (AI) could be trained to perform anticipated psychotherapy health services with reliable conversation output and fast learning behavior. Building a generated pre-trained transformer, (GPT) large language model to train in mental health with a deep understanding of user experience could make providing an appropriate level of care with AI conversation possible, as different models each have their individual areas of focus. Dominating technology companies, like NVIDIA, offer a customization experience that "...lets you personalize a GPT large language model (LLM) connected to your own content—docs, notes, images, or other data." (Build a Custom LLM with ChatRTX | NVIDIA, n.d.). A chat model therefore could realistically be grounded in real treatment documentation kept private through a careful approach to data access and handling. Along with grounding references to published mental health research would make a subject matter expert out of the healthcare GPT. GPT communication could progress to be aligned to mimic a talk therapy experience and could be overseen by a licensed

psychotherapist, allowing for increased productivity. More time devoted to tasks that serve more people would allow a broader reach of patients to treat according to the analysis generated from each chat, or therapy, session. There are shared beliefs people are already expecting progress like this to become a new normal. A United States government user research experiment concluded that “all of our participants had used web chat before. They see it as an increasingly common feature for interacting with retailers and service providers.” (USAGov’s Chatbot Users Talk Back: Find Out What They’re Teaching Us | USAGov, n.d.). Consumers from the user research shared they usually anticipate ability to escalate AI interactions to a human interaction where the psychotherapist AI observational role could include therapists’ occasional manual treatment interaction. As of January 28, 2026, OpenAI received “...more than 1,000 high-quality entries from both established and emerging researchers from around the world, making this one of our largest calls for research grants to date” (Funding Grants for New Research into AI and Mental Health | OpenAI, n.d.) where up to \$2 million has been designated to fund independent safety and well-being research. Proposals have already received awards for health-related research with OpenAI. In summary, OpenAI and companies like it could expand the scope of patients treated with insurance billings per healthcare provider. This would be made possible by specifying LLMs trained on documented mental health treatment services and published psychological and psychiatric studies to provide the appropriate psychotherapy service based on individual user experience. Trust could maybe build the future by allowing AI to prescribe medicine.

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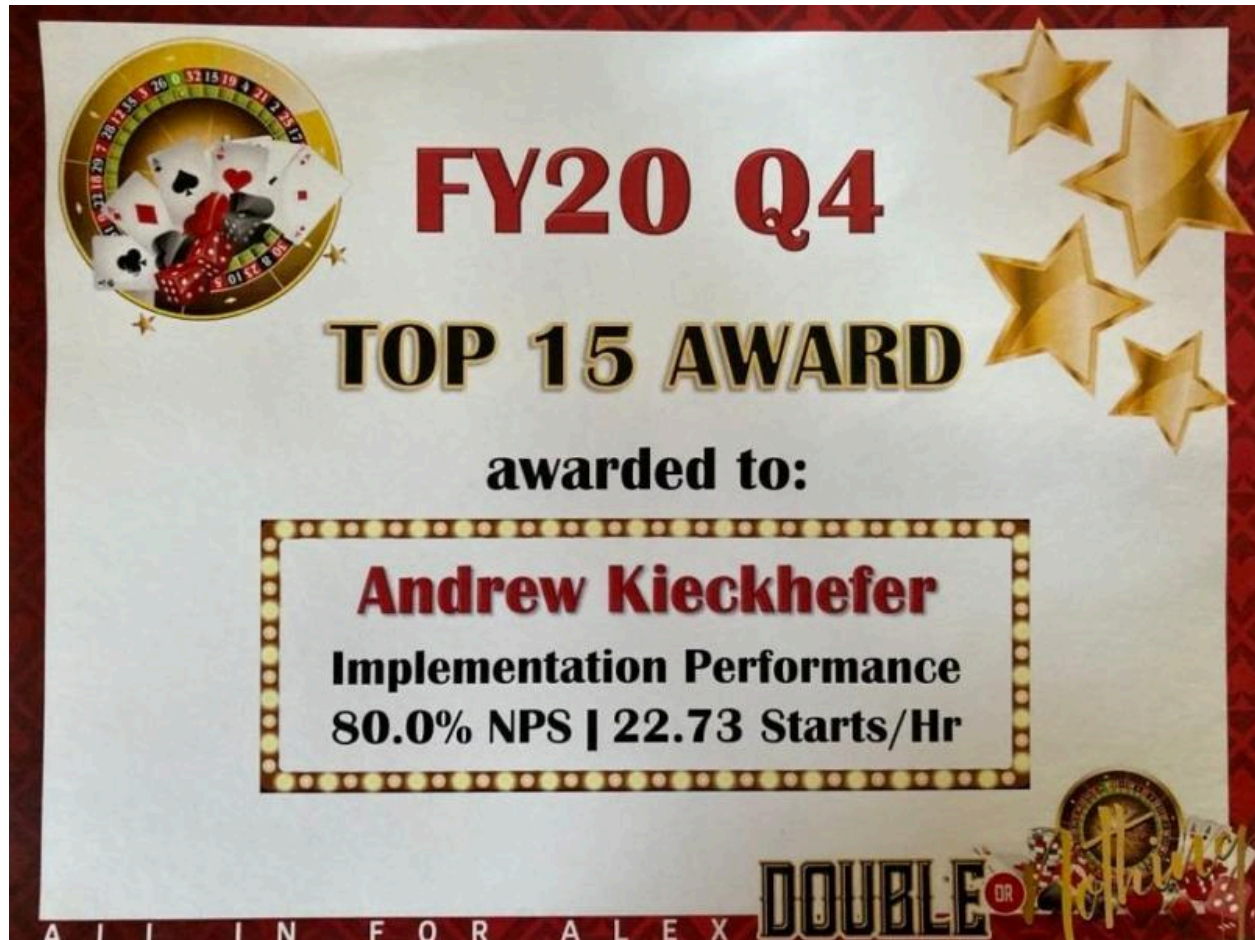
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# ADP WORKPLACE EFFICIENCY



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**Andrew Kieckhefer**

*Curriculum Vitae*

*Thank You For Reading*